ON THE KELLY MONOIDAL STRUCTURE OF Λ-SEQUENCES

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Abstract. Let Λ be the category of based finite sets **n** and based injections. We study properties of monoids and modules in Λ-sequences under the Kelly monoidal structure. In particular, we show that the forgetful functor from right modules in Λ-sequences to right modules in symmetric sequences is an isomorphism. We show that any compatible lower data extends to a normal oplax monoidal structure and use this to establish a universal normal oplax monoidal structure on Λ-sequences extending the Kelly product, identifying unital operads to monoids in unital Λ-sequences for a general symmetric monoidal category $\mathscr{V}$. We also establish a closed monoidal localization theorem.

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Introduction

The notion of operads was introduced by Peter May in the 1970s [May72], and has since been prevalent in studying algebraic structures in the context of topological spaces, spectra, algebras and more. It was first observed by Kelly [Kel05] that the structure of an operad in a closed symmetric monoidal category $\mathscr{V}$ is the same as

a monoid in symmetric sequences in $\mathscr{V}$, under a product $\odot_\Sigma$ he defined. This product is also called the composition product (see [MZZ25] for a historical account). This perspective is convenient for studying model structures on operads and their algebras, or for forming bar constructions [BM03, BM06, Kro07, MZZ25, PS18].

An operad $\mathscr{C}$ in a symmetric monoidal category $(\mathscr{V}, \otimes, \mathcal{I})$ is called *based* if it comes with a map $\eta : \mathcal{I} \to \mathscr{C}(0)$ and *unital* if $\mathscr{C}(0) = \mathcal{I}$. Unital operads are relevant in the context of based spaces, spectra or unital k-algebras, as they build in the pre-assigned based condition of algebras into the combinatorics of the operad. In [MZZ25], May, Zhang and the second named author developed a unital version of Kelly's theorem: replacing symmetric sequences by Λ-sequences, unital (resp. based) operads in $\mathscr{V}$ can be identified with monoids in unital Λ-sequences (resp. Λ-sequences) in $\mathscr{V}$ under the Kelly product $\odot_\Lambda$ (see Section 1 for definitions).

Modules. The identification of operads as monoids opens up the perspective of left, right modules and bar constructions. In Section 2, we study properties of monoids and modules in Σ-objects and Λ-objects in a closed symmetric monoidal category $\mathscr{V}$. In particular, we show the following result in Theorem 2.6.

Theorem 0.1. *Let $\mathscr{C}$ be a based operad. Then the forgetful map from right $\mathscr{C}$-modules in Λ-objects to right $\mathscr{C}$-modules in Σ-objects is an isomorphism.*

Some special cases of this results are already known. Taking $\mathscr{C} = \mathscr{I}_1$ where $\mathscr{I}_1(\mathbf{0}) = \mathscr{I}_1(\mathbf{1}) = \mathcal{I}$ and $\mathscr{I}_1(\mathbf{n}) = \emptyset$ for $n \geq 2$, this is [Fre09, 5.1.8]; taking $\mathscr{C}$ to be the commutativity operad, this is [MZZ25, 4.10] and [Fre09, 5.1.6]. See Example 2.34.

We also find an interesting construction of new operads. For an operad $\mathscr{C}$ in a Cartesian monoidal category $\mathscr{V}$, we construct a new operad $\mathscr{B}$ with explicit terms

$$\begin{aligned} \mathscr{B}(0) &= \mathscr{C}(0), \\ \mathscr{B}(1) &= \mathscr{C}(0) \sqcup \mathscr{C}(1), \\ \text{and } \mathscr{B}(n) &= \sqcup_{S \subset \{1,2,\cdots,n\}} \mathscr{C}(|S|). \end{aligned}$$

The category of $\mathscr{B}$-algebras seems ridiculously rich: each $\mathscr{B}$-algebra is simultaneously a $\mathscr{C}$-algebra and a commutative algebra; both $\mathscr{C}$-algebras and commutative algebras are faithful subcategories of $\mathscr{B}$-algebras (see Proposition 2.14 and remarks after). We also speculate that $\mathscr{B}$ has an interesting role in the infinite loop space theory (see Remark 2.27).

From an operad $\mathscr{C}$ in $\mathscr{V}$, two associated $\mathscr{V}$-enriched symmetric monoidal categories have been studied in the literature: the permutative envelop $\overline{\mathscr{C}}$ and the category of operators $\widehat{\mathscr{C}}$. They are constructed in explicit terms. In [MZZ25], it is shown how to construct $\overline{\mathscr{C}}$ using the Day convolution and Kelly product; in Proposition 2.24, we show how to construct $\widehat{\mathscr{C}}$ using the Day convolution and Kelly product.

The permutative envelop $\overline{\mathscr{C}}$ has the universal property that the category of $\mathscr{C}$-algebras in $\mathscr{V}$ is isomorphic to strong symmetric monoidal functors from $\overline{\mathscr{C}}$ to $\mathscr{V}$. When $\mathscr{V}$ is Cartesian monoidal, this is also isomorphic to strong symmetric monoidal functors from $\widehat{\mathscr{C}}$ to $\mathscr{V}$. However, in homotopy theory it is common to replace the strong symmetric monoidal condition with the Segal conditions, then $\widehat{\mathscr{C}}$ becomes more interesting. For a subcategory $\Pi \subset \widehat{\mathscr{C}}$, May–Thomason [MT78] uses a monad $\widehat{\mathbb{C}}^{op}_\Pi$ on $\Pi[\mathrm{Top}_*]$ to unify Segal's $\mathscr{F}$-space approach and May's operadic

approach to infinite loop space machines. Here, $\Pi[\mathrm{Top}_*]$ is the category of covariant functors from Π to based spaces.

We show in Lemma 2.32 that for $\Sigma \subset \Lambda \subset \overline{\mathscr{C}}$, the monads $\overline{\mathbb{C}}_\Sigma^{op}$ and $\overline{\mathbb{C}}_\Lambda^{op}$ are isomorphic to the monads associated to the monoid $\mathscr{C}$ in Σ-objects and Λ-objects. With this we show the following result in Theorem 2.33 for both Σ-objects and Λ-objects, which gives a complete understanding of right $\mathscr{C}$-modules. Note that $\overline{\mathscr{C}}^{op}$ does not depend on Σ or Λ, so this also gives another proof of Theorem 0.1.

Theorem 0.2. *The category of right $\mathscr{C}$-modules is isomorphic to the category of $\mathscr{V}$-enriched functors from $\overline{\mathscr{C}}^{op}$ to $\mathscr{V}$.*

Unital operads as monoids. In order for the Kelly product to be a monoidal product, one uses that the monoidal structure on $\mathscr{V}$ is closed and in particular that $\otimes$ commutes with small colimits. Sometimes one would like to drop this assumption. For example, Top^{op} is not closed monoidal and the operads in it are cooperads in Top.

In [Chi12, Definitions 2.10-2.12], Ching generalized the notion of a monoidal structure on a category $\mathscr{E}$ to a so-called normal oplax monoidal structure, namely for all $n \geq 2$ functors

$$\mu_n : \mathscr{E}^n \to \mathscr{E},$$

and some compatible natural transformations $\alpha_{n,l,r}$. He then constructed by hand a normal oplax monoidal structure on symmetric sequences in $\mathscr{V}$ where μ_2 is the Kelly product. This way, he could still make sense of monoids and bar constructions. It is left open in [MZZ25] whether this can be generalized to Λ-sequences. In fact, this approach could work, with significant modification due to the base point identifications (Remark 4.16).

In Section 4, we offer an alternative approach by studying the universal case in general context. First, we show:

Theorem 0.3. *(Theorem 4.30) In a category $\mathscr{E}$ with small limits, any compatible data of μ_n and $\alpha_{n,l,r}$ for $n \leq 3$ extends to normal oplax structures, and there is a universal one.*

Then, using a monoidal localization theorem we established, we prove:

Theorem 0.4. *(Theorem 4.26) There is a universal normal oplax monoidal structure on Λ-sequences where μ_2 and μ_3 are the 2-fold and 3-fold Kelly products.*

In particular, the adaption of Ching's normal oplax monoidal structure has a morphism to our construction. Moreover, all sensible normal oplax monoidal structures are equivalent, having the same category of monoids (Remark 4.33). Using this structure, we establish the following result in Theorem 4.32, answering a question left open in [MZZ25].

Theorem 0.5. *Let $\mathscr{V}$ be a complete and cocomplete symmetric monoidal category. There are isomorphisms between categories of unital operads in $\mathscr{V}$ and of monoids in unital Λ-sequences in $\mathscr{V}$; between categories of based operads in $\mathscr{V}$ and of monoids in Λ-sequences in $\mathscr{V}$.*

Monoidal localization. We study the localization of categories in Section 3. As a key step of our proof of Theorem 0.4, we establish the following *closed monoidal localization theorem* (Theorem 3.9), which could be of independent interest.

Theorem 0.6. *Let $\mathscr{V}$ be a complete and cocomplete symmetric monoidal category. There exists a localization $L : \mathscr{V} \to L\mathscr{V} = \mathscr{V}[\overline{S}^{-1}]$ such that*

- *$L\mathscr{V}$ is complete, cocomplete and symmetric monoidal;*
- *L preserves small limits, colimits and $\otimes$;*
- *$\otimes$ commutes with small colimits in $L\mathscr{V}$.*

Remark 0.7. If $L\mathscr{V}$ is presentable, it is closed monoidal by the adjoint functor theorem.

Day showed in [Day73, Proposition 2.5] that any monoidal category can be embedded into a complete and cocomplete category in a way that preserves both the monoidal structure, limits and those colimits *that are respected by the tensor product*. The goal of his localization is for cocompleteness while the goal of our localization is to commute $\otimes$ with colimits.

In the same paper, Day introduced the concept of a *monoidal* collection of morphisms S in $\mathscr{V}$ such that the localization $L : \mathscr{V} \to \mathscr{V}[S^{-1}]$ preserves the tensor product. We introduce the concept of S being a *κ-closed monoidal multiplicative system*, so that the localization $L : \mathscr{V} \to \mathscr{V}[S^{-1}]$ preserves the tensor product as well as κ-small limits and colimits. We then show in Theorem 3.7 that

Theorem 0.8. *Let κ be a regular cardinal. Any collection of morphisms S has a κ-closed monoidal multiplicative closure $\overline{S}$.*

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1. Preliminaries

We recall some definitions and results from [MZZ25]. Let $\mathscr{V}$ be a complete and cocomplete symmetric monoidal category throughout. The category $\mathscr{V}$ has three distinguished objects:

(1) $\emptyset$, the initial object, that is, the coproduct of the empty set of objects.
(2) $*$, the terminal object, that is, the product of the empty set of objects.
(3) $\mathcal{I}$, the unit object for $\otimes$ on $\mathscr{V}$.

Definition 1.1. The category Λ has objects of based finite sets $\mathbf{n} = \{0, 1, \cdots, n\}$, with base point 0, and morphisms of based injections. The category Σ is the subcategory of Λ with the same objects and isomorphisms, or permutations. Note that morphisms in Λ are generated by the permutations and ordered injections $\sigma_i \colon \mathbf{n} - \mathbf{1} \to \mathbf{n}$, $1 \leq i \leq n$, that skip i in the target.

Notation 1.2. A *symmetric sequence* in $\mathscr{V}$ is a functor $\Sigma^{op} \to \mathscr{V}$; a *(unitary) Λ-sequence* in $\mathscr{V}$ is a functor $\mathscr{D} : \Lambda^{op} \to \mathscr{V}$ together with a base map $\eta \colon \mathcal{I} \to \mathscr{D}(\mathbf{0})$. We say that a Λ-sequence $\mathscr{D}$ is *unital* if $\mathscr{D}(\mathbf{0}) = \mathcal{I}$ and η is the identity map. We write $\Sigma^{op}[\mathscr{V}]$, $\Lambda^{op}[\mathscr{V}]$, $\Lambda^{op}[\mathscr{V}]_{\mathscr{I}_0}$ and $\Lambda^{op}_{\mathcal{I}}[\mathscr{V}]$ for the category of Σ-sequences, contravariant Λ-functors, Λ-sequences and unital Λ-sequences, where the morphisms are natural transformations (that restrict to morphisms under $\mathcal{I}$ at $\mathbf{0}$). We have an inclusion of categories $\Lambda^{op}_{\mathcal{I}}[\mathscr{V}] \subset \Lambda^{op}[\mathscr{V}]_{\mathscr{I}_0}$.

Definition 1.3. For $m \geq 0$, we let $\mathscr{I}_m$ denote the Λ-sequence

$$\mathscr{I}_m = \Lambda(-, \mathbf{m}) \otimes \mathcal{I}.$$

In particular, $\mathscr{I}_0(\mathbf{0}) = \mathcal{I}$ and $\mathscr{I}_0(\mathbf{n}) = \emptyset$ for $n \geq 1$; $\mathscr{I}_1(\mathbf{0}) = \mathscr{I}_1(\mathbf{1}) = \mathcal{I}$ and $\mathscr{I}_1(\mathbf{n}) = \emptyset$ for $n \geq 2$. We abuse notation to also write the base map as $\eta : \mathscr{I}_0 \to \mathscr{D}$.

For $\mathscr{D}$ and $\mathscr{E}$ in $\Lambda^{op}[\mathscr{V}]$, the Day convolution $\mathscr{D}\boxtimes\mathscr{E}$ is the left Kan extension displayed in the diagram

$$\begin{array}{ccccc} \Lambda^{op}\times\Lambda^{op} & \xrightarrow{\mathscr{D}\times\mathscr{E}} & \mathscr{V}\times\mathscr{V} & \xrightarrow{\otimes} & \mathscr{V}. \\ \vee\downarrow & & & \nearrow_{\mathscr{D}\boxtimes\mathscr{E}} & \\ \Lambda^{op} & & & & \end{array} \tag{1.4}$$

It can be computed by a coend

$$(\mathscr{D}\boxtimes\mathscr{E})(\mathbf{n}) = \int^{\mathbf{j}_1,\mathbf{j}_2} \Lambda(\mathbf{n},\mathbf{j_1}+\mathbf{j_2})\otimes\big(\mathscr{D}(\mathbf{j}_1)\otimes\mathscr{E}(\mathbf{j}_2)\big). \tag{1.5}$$

As proved in [Kel74, 2.1], it is formal that $\boxtimes$ is a symmetric monoidal product which is closed when $\mathscr{V}$ is and that $\mathscr{I}_0$ is the unit for the Day convolution.

When $\mathscr{D}$ and $\mathscr{E}$ are Λ-sequences, so is $\mathscr{D}\boxtimes\mathscr{E}$. The base maps of $\mathscr{D}$ and $\mathscr{E}$ induce the base map of $\mathscr{D}\boxtimes\mathscr{E}$ by

$$\eta : \mathscr{I}_0 \cong \mathscr{I}_0\boxtimes\mathscr{I}_0 \xrightarrow{\eta\boxtimes\eta} \mathscr{D}\boxtimes\mathscr{E}.$$

One can perform the Day convolution on $\Sigma^{op}[\mathscr{V}]$ similarly, and we write $\boxtimes_\Lambda$ and $\boxtimes_\Sigma$ to indicate the context. For $\mathscr{D}$ and $\mathscr{E}$ in $\Lambda^{op}[\mathscr{V}]$, there is a natural map of Σ-sequences $q\colon \mathscr{D}\boxtimes_\Sigma\mathscr{E}\to\mathscr{D}\boxtimes_\Lambda\mathscr{E}$, which turns out to be an isomorphism of Σ-sequences [MZZ25, Theorem 3.4]. So we may write $\boxtimes$ for both $\boxtimes_\Lambda$ and $\boxtimes_\Sigma$. Explicitly,

$$(\mathscr{D}\boxtimes\mathscr{E})(\mathbf{n}) \cong \coprod_{j+k=n} \mathscr{D}(\mathbf{j})\otimes\mathscr{E}(\mathbf{k})\otimes_{\Sigma_j\times\Sigma_k}\Sigma_n.$$

Remark 1.6. One has the Day convolution product for covariant sequences as well. For $\mathscr{C},\mathscr{D}\in\Lambda[\mathscr{V}]$, the natural map $q\colon \mathscr{D}\boxtimes_\Sigma\mathscr{E}\to\mathscr{D}\boxtimes_\Lambda\mathscr{E}$ is no longer an isomorphism: For example, in $(\mathscr{C}\boxtimes_\Lambda\mathscr{D})(\mathbf{3})$, there is one copy of $\mathscr{C}(3)\otimes\mathscr{D}(0)$ and three copies of $\mathscr{C}(2)\otimes\mathscr{D}(0)$. All three copies are glued to $\mathscr{C}(3)\otimes\mathscr{D}(0)$ via different maps, yielding a quotient space of $(\mathscr{C}\boxtimes_\Sigma\mathscr{D})(\mathbf{3})$.

When $\otimes$ commutes with finite colimits in $\mathscr{V}$, the Day convolution is associative, and there is

$$\mathscr{D}_1\boxtimes_\Lambda\cdots\boxtimes_\Lambda\mathscr{D}_n = \int^{i_1,\dots,i_n} \Lambda(-,\mathbf{i_1}+\cdots+\mathbf{i_n})\otimes\Big(\mathscr{D}_1(\mathbf{i_1})\otimes\cdots\otimes\mathscr{D}_n(\mathbf{i_n})\Big) \tag{1.7}$$

Definition 1.8. Define the Kelly product $\odot_\Lambda$ on $\Lambda^{op}[\mathscr{V}]_{\mathscr{I}_0}$ by

$$\big(\mathscr{D}\odot_\Lambda\mathscr{E}\big)(\mathbf{n}) = \int^{k} \mathscr{D}(\mathbf{k})\otimes\mathscr{E}^{\boxtimes k}(\mathbf{n}).$$

The base maps of $\mathscr{D}$ and $\mathscr{E}$ induce the base map of $\mathscr{D}\odot_\Lambda\mathscr{E}$ as the composite

$$\mathscr{I}_0\xrightarrow{\eta}\mathscr{D}=\mathscr{D}\odot_\Lambda\mathscr{I}_0\xrightarrow{\mathscr{D}\odot_\Lambda\eta}\mathscr{D}\odot_\Lambda\mathscr{E}.$$

Observe that if $\mathscr{D}(\mathbf{0})=\mathcal{I}$ and $\mathscr{E}(\mathbf{0})=\mathcal{I}$, then these are identity maps, so that $\odot$ restricts to a product on unital Λ-sequences.

The Kelly product $\odot_\Sigma$ on $\Sigma^{op}[\mathscr{V}]$ is defined by replacing Λ by Σ and ignoring the base maps in this definition.

Remark 1.9. The Kelly product is also defined when the first variable does not have a base map, as a functor

$$\odot_\Lambda : \Lambda^{op}[\mathscr{V}] \times \Lambda^{op}[\mathscr{V}]_{\mathscr{I}_0} \to \Lambda^{op}[\mathscr{V}].$$

When $\mathscr{V}$ is closed symmetric monoidal, we have:

Theorem 1.10. *([MZZ25, Theorem 4.5]) The category $\Lambda^{op}[\mathscr{V}]_{\mathscr{I}_0}$ is monoidal with the Kelly product $\odot_\Lambda$ and unit object $\mathscr{I}_1$.*

Definition 1.11. A monoid in $\Lambda^{op}[\mathscr{V}]_{\mathscr{I}_0}$ is an object $\mathscr{C}$ equipped with a unit map $\eta_\times : \mathscr{I}_1 \to \mathscr{C}$ and a product map $\gamma : \mathscr{C} \odot_\Lambda \mathscr{C} \to \mathscr{C}$ such that the following diagrams commute:

$$\begin{array}{ccccc} \mathscr{I}_1 \odot_\Lambda \mathscr{C} & \xrightarrow{\eta_\times \odot_\Lambda \mathrm{id}} & \mathscr{C} \odot_\Lambda \mathscr{C} & \xleftarrow{\mathrm{id} \odot_\Lambda \eta_\times} & \mathscr{C} \odot_\Lambda \mathscr{I}_1 \\ & \searrow^{\cong} & \downarrow{\scriptstyle \gamma} & \swarrow_{\cong} & \\ & & \mathscr{C} & & \end{array} \qquad \begin{array}{ccc} \mathscr{C} \odot_\Lambda \mathscr{C} \odot_\Lambda \mathscr{C} & \xrightarrow{\gamma \odot_\Lambda \mathrm{id}} & \mathscr{C} \odot_\Lambda \mathscr{C} \\ \downarrow{\scriptstyle \mathrm{id} \odot_\Lambda \gamma} & & \downarrow{\scriptstyle \gamma} \\ \mathscr{C} \odot_\Lambda \mathscr{C} & \xrightarrow[\gamma]{} & \mathscr{C} \end{array}$$

A right module over a monoid $\mathscr{C}$ in $\Lambda^{op}[\mathscr{V}]_{\mathscr{I}_0}$ is an object $\mathscr{M}$ with a right action map $\rho : \mathscr{M} \odot_\Lambda \mathscr{C} \to \mathscr{M}$ such that the following diagrams commute:

$$\begin{array}{ccc} \mathscr{M} \odot_\Lambda \mathscr{C} & \xleftarrow{\mathrm{id} \odot_\Lambda \eta_\times} & \mathscr{M} \odot_\Lambda \mathscr{I}_1 \\ \downarrow{\scriptstyle \rho} & \swarrow_{\cong} & \\ \mathscr{M} & & \end{array} \qquad \begin{array}{ccc} \mathscr{M} \odot_\Lambda \mathscr{C} \odot_\Lambda \mathscr{C} & \xrightarrow{\rho \odot_\Lambda \mathrm{id}} & \mathscr{M} \odot_\Lambda \mathscr{C} \\ \downarrow{\scriptstyle \mathrm{id} \odot_\Lambda \gamma} & & \downarrow{\scriptstyle \rho} \\ \mathscr{M} \odot_\Lambda \mathscr{C} & \xrightarrow[\rho]{} & \mathscr{M} \end{array}$$

A left module over $\mathscr{C}$ is an object $\mathscr{M}$ with a left action map $\theta : \mathscr{C} \odot_\Lambda \mathscr{M} \to \mathscr{M}$ such that similar diagrams as above where $\mathscr{M}$ is placed in the right instead of in the left commute.

Remark 1.12. A monoid $\mathscr{C}$ has two "unit" maps, the additive one $\eta_+ : \mathscr{I}_0 \to \mathscr{C}$ (the base map, denoted as η before) and the multiplicative one $\eta_\times : \mathscr{I}_1 \to \mathscr{C}$. For brevity, we will simply denote them by η when there is no confusion.

2. Modules

In this section, we always assume that $(\mathscr{V}, \otimes, \mathcal{I})$ is a closed complete and cocomplete symmetric monoidal category.

2.1. **Monoids and modules.** The following results established the Kelly product as a monoidal product and identified the corresponding monoids. Recall that $\mathscr{I}_1(\mathbf{0}) = \mathscr{I}_1(\mathbf{1}) = \mathcal{I}$ and $\mathscr{I}_1(\mathbf{n}) = \emptyset$ for $n \geq 2$. Let $\mathscr{I}_1^\Sigma \in \Sigma^{op}[\mathscr{V}]$ be such that $\mathscr{I}_1(\mathbf{1}) = \mathcal{I}$ and $\mathscr{I}_1(\mathbf{n}) = \emptyset$ for $n \neq 1$.

Theorem 2.1. *We have the following results on Kelly products:*

1. *The Kelly product $\odot_\Sigma$ is associtaive with unit $\mathscr{I}_1^\Sigma$ and strong monoidal in the first variable.*
2. *Operads in $\mathscr{V}$ are the same as monoids in $(\Sigma^{op}[\mathscr{V}], \odot_\Sigma, \mathscr{I}_1^\Sigma)$.*
3. *The Kelly product $\odot_\Lambda$ is associtaive with unit $\mathscr{I}_1$ and strong monoidal in the first variable.*
4. *Unital operads in $\mathscr{V}$ are the same as monoids in $(\Lambda_{\mathcal{I}}^{op}[\mathscr{V}], \odot_\Lambda, \mathscr{I}_1)$ and based operads in $\mathscr{V}$ are the same as monoids in $(\Lambda^{op}[\mathscr{V}]_{\mathscr{I}_0}, \odot_\Lambda, \mathscr{I}_1)$.*

(5) *If $\mathscr{V}$ is Cartesian monoidal, the Kelly products $\odot_\Sigma$ and $\odot_\Lambda$ are oplax monoidal in the first variable.*

Proof. Properties (1)-(4) are proved in [Kel05, MZZ25]. For (5), note that when $\mathscr{V}$ is Cartesian monoidal, we have diagonal morphisms $\Delta : A \to A \otimes A$ for $A \in \mathscr{V}$. Then we have

$$
\begin{aligned}
&\mathscr{E} \odot_\Lambda (\mathscr{D} \boxtimes \mathscr{D}')(\mathbf{n}) \\
&= \int^{\mathbf{m}} \mathscr{E}(\mathbf{m}) \otimes (\mathscr{D} \boxtimes \mathscr{D}')^{\boxtimes m}(\mathbf{n}) \\
&\cong \int^{\mathbf{m}} \mathscr{E}(\mathbf{m}) \otimes \big(\mathscr{D}^{\boxtimes m} \boxtimes \mathscr{D}'^{\boxtimes m}\big)(\mathbf{n}) \\
&\to \int^{\mathbf{m}} \mathscr{E}(\mathbf{m}) \otimes \mathscr{E}(\mathbf{m}) \otimes \big(\mathscr{D}^{\boxtimes m} \boxtimes \mathscr{D}'^{\boxtimes m}\big)(\mathbf{n}) \\
&= \int^{\mathbf{m}} \mathscr{E}(\mathbf{m}) \otimes \mathscr{E}(\mathbf{m}) \otimes \left(\int^{\mathbf{n_1},\mathbf{n_2}} \Lambda(\mathbf{n}, \mathbf{n}_1 + \mathbf{n}_2) \otimes \mathscr{D}^{\boxtimes m}(\mathbf{n}_1) \otimes \mathscr{D}'^{\boxtimes m}(\mathbf{n}_2) \right) \\
&\cong \int^{\mathbf{n_1},\mathbf{n_2},\mathbf{m}} \Lambda(\mathbf{n}, \mathbf{n}_1 + \mathbf{n}_2) \otimes \mathscr{E}(\mathbf{m}) \otimes \mathscr{D}^{\boxtimes m}(\mathbf{n}_1) \otimes \mathscr{E}(\mathbf{m}) \otimes \mathscr{D}'^{\boxtimes m}(\mathbf{n}_2) \\
&\xrightarrow{\delta_*} \int^{\mathbf{n_1},\mathbf{n_2},\mathbf{m_1},\mathbf{m_2}} \Lambda(\mathbf{n}, \mathbf{n}_1 + \mathbf{n}_2) \otimes \mathscr{E}(\mathbf{m_1}) \otimes \mathscr{D}^{\boxtimes m_1}(\mathbf{n}_1) \otimes \mathscr{E}(\mathbf{m_2}) \otimes \mathscr{D}'^{\boxtimes m_2}(\mathbf{n}_2) \\
&\cong \int^{\mathbf{n_1},\mathbf{n_2}} \Lambda(\mathbf{n}, \mathbf{n}_1 + \mathbf{n}_2) \otimes \Big(\int^{\mathbf{m_1}} \mathscr{E}(\mathbf{m_1}) \otimes \mathscr{D}^{\boxtimes m_1}(\mathbf{n}_1) \Big) \otimes \Big(\int^{\mathbf{m_2}} \mathscr{E}(\mathbf{m_2}) \otimes \mathscr{D}'^{\boxtimes m_2}(\mathbf{n}_2) \Big) \\
&= (\mathscr{E} \odot_\Lambda \mathscr{D}) \boxtimes (\mathscr{E} \odot_\Lambda \mathscr{D}')(\mathbf{n}).
\end{aligned}
$$

Here, the equalities are denifitions of $\boxtimes$ and $\odot_\Lambda$; the first isomorphism uses that $\boxtimes$ is symmetric monoidal; the arrow afterwards is induced by the diagonal morphisms Δ; the remaining two isomorphisms commute colimits with $\otimes$; and the arrow labelled by δ_* is a comparison of colimits of diagrams indexed along the diagonal map $\delta : \Lambda \to \Lambda \times \Lambda$. The proof also works for $\odot_\Sigma$ replacing all Λ by Σ. □

We list some consequences of this theorem for both contexts of Σ-sequences and Λ-sequences.

Proposition 2.2. *If $\mathscr{E}$ is a right $\mathscr{C}$-module, then $\mathscr{D} \odot \mathscr{E}$ is a right $\mathscr{C}$-module. If $\mathscr{D}$ is a left $\mathscr{C}$-module, then $\mathscr{D} \odot \mathscr{E}$ is a left $\mathscr{C}$-module.*

Proof. For the first case, the right $\mathscr{C}$-action on $\mathscr{D} \odot \mathscr{E}$ is given by

$$(\mathscr{D} \odot \mathscr{E}) \odot \mathscr{C} \cong \mathscr{D} \odot (\mathscr{E} \odot \mathscr{C}) \xrightarrow{\mathrm{id} \odot \rho_{\mathscr{E}}} \mathscr{D} \odot \mathscr{E}.$$

The second case is similar. □

Proposition 2.3. *If $\mathscr{D}$ and $\mathscr{E}$ are right $\mathscr{C}$-modules, then $\mathscr{D} \boxtimes \mathscr{E}$ is a right $\mathscr{C}$-module.*

Proof. The right $\mathscr{C}$-action on $\mathscr{D} \boxtimes \mathscr{E}$ is given by

$$(\mathscr{D} \boxtimes \mathscr{E}) \odot \mathscr{C} \cong (\mathscr{D} \odot \mathscr{C}) \boxtimes (\mathscr{E} \odot \mathscr{C}) \xrightarrow{\rho_{\mathscr{D}} \boxtimes \rho_{\mathscr{E}}} \mathscr{D} \boxtimes \mathscr{E},$$

where the isomorphism uses Theorem 2.1(1)(3). □

Proposition 2.4. *If $\mathscr{D}$ and $\mathscr{E}$ are left $\mathscr{C}$-modules in a Cartesion monoidal category $\mathscr{V}$, then $\mathscr{D} \boxtimes \mathscr{E}$ is a left $\mathscr{C}$-module.*

Proof. The left $\mathscr{C}$-action on $\mathscr{D} \boxtimes \mathscr{E}$ is given by

$$\mathscr{C} \odot (\mathscr{D} \boxtimes \mathscr{E}) \longrightarrow (\mathscr{C} \odot \mathscr{D}) \boxtimes (\mathscr{C} \odot \mathscr{E}) \xrightarrow{\theta_{\mathscr{D}} \boxtimes \theta_{\mathscr{E}}} \mathscr{D} \boxtimes \mathscr{E},$$

where the first map uses Theorem 2.1(5). □

2.2. **Comparing Λ-sequences and Σ-sequences.** Any Λ-sequence can be viewed as a Σ-sequence by forgetting structures.

Proposition 2.5. *The forgetful map restricts to the catogories of monoids as well as to the categories of modules.*

Proof. We denote by $\iota : \mathscr{I}_1^{\Sigma} \to \mathscr{I}_1$ the inclusion map of Σ-sequences. For $\mathscr{C}, \mathscr{D}$ in $\Lambda^{op}[\mathscr{V}]_{\mathscr{I}_0}$, there is a canonical quotient map $q : \mathscr{C} \odot_{\Sigma} \mathscr{D} \to \mathscr{C} \odot_{\Lambda} \mathscr{D}$ of Σ-sequences. The two diagrams

$$\begin{array}{ccc} \mathscr{I}_1^{\Sigma} \odot_{\Sigma} \mathscr{D} & \xrightarrow{\cong} & \mathscr{D} \\ \downarrow{\scriptstyle \iota} & & \| \\ \mathscr{I}_1 \odot_{\Sigma} \mathscr{D} & & \| \\ \downarrow{\scriptstyle q} & & \| \\ \mathscr{I}_1 \odot_{\Lambda} \mathscr{D} & \xrightarrow{\cong} & \mathscr{D} \end{array} \qquad \begin{array}{ccc} \mathscr{D} \odot_{\Sigma} \mathscr{I}_1^{\Sigma} & \xrightarrow{\cong} & \mathscr{D} \\ \downarrow{\scriptstyle \iota} & & \| \\ \mathscr{D} \odot_{\Sigma} \mathscr{I}_1 & & \| \\ \downarrow{\scriptstyle q} & & \| \\ \mathscr{D} \odot_{\Lambda} \mathscr{I}_1 & \xrightarrow{\cong} & \mathscr{D} \end{array}$$

comparing the unit maps commute. Via the maps q and ι, the forgetful map restricts to

$$\mathrm{fgt} : \mathrm{Mon}(\Lambda^{op}[\mathscr{V}]_{\mathscr{I}_0}) \to \mathrm{Mon}(\Sigma^{op}[\mathscr{V}])$$

as well as to the categories of modules. □

Theorem 2.6. *Let $\mathscr{C}$ be a monoid in $(\Lambda^{op}[\mathscr{V}]_{\mathscr{I}_0}, \odot_{\Lambda})$, i.e. a based operad. Then the forgetful map induces*

$$\mathrm{RMod}_{\mathscr{C}}[\Lambda^{op}[\mathscr{V}]] \cong \mathrm{RMod}_{\mathscr{C}}[\Sigma^{op}[\mathscr{V}]].$$

In particular, $\Lambda^{op}[\mathscr{V}] \cong \mathrm{RMod}_{\mathscr{I}_1}[\Lambda^{op}[\mathscr{V}]] \cong \mathrm{RMod}_{\mathscr{I}_1}[\Sigma^{op}[\mathscr{V}]]$.

Proof. An object $\mathscr{D}$ in $\mathrm{RMod}_{\mathscr{C}}[\Sigma^{op}[\mathscr{V}]]$ is a morphism $\mathscr{D} \odot_{\Sigma} \mathscr{C} \to \mathscr{D}$ satisfying compatibility conditions. Explicitly, for $n \geqslant 0$ there are morphisms

$$\coprod_{k, n_1, \cdots, n_k} \coprod_{\Sigma_n} \mathscr{D}(\mathbf{k}) \otimes \mathscr{C}(\mathbf{n_1}) \otimes \cdots \mathscr{C}(\mathbf{n_k}) \to \mathscr{D}(\mathbf{n}), \tag{2.7}$$

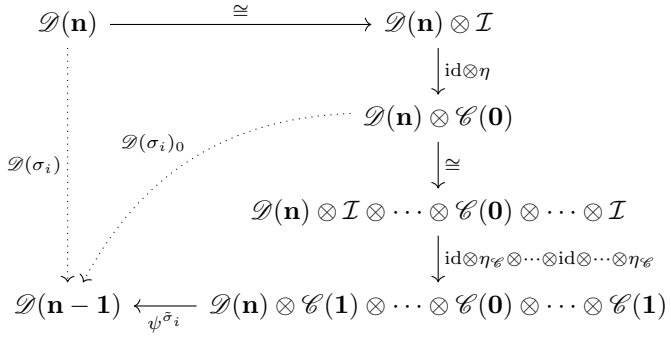
which is equivalent to a series of morphisms

$$\psi^{\sigma}_{k, n_1, \cdots, n_k} : \mathscr{D}(\mathbf{k}) \otimes \mathscr{C}(\mathbf{n_1}) \otimes \cdots \mathscr{C}(\mathbf{n_k}) \to \mathscr{D}(\mathbf{n}) \tag{2.8}$$

where $n = n_1 + \cdots + n_k; \sigma \in \Sigma(\mathbf{n}, \mathbf{n_1} + \cdots + \mathbf{n_k}) = \Sigma_n$.

We first prove that the family of morphisms $\{\phi^{\sigma}_{k, n_1, \cdots, n_k}\}$ determines a Λ-sequence structure on $\mathscr{D}$. For $\sigma_i \in \Lambda(\mathbf{n-1}, \mathbf{n})$, we denote $\tilde{\sigma_i}$ to be the corresponding element in $\Lambda(\mathbf{n-1}, \mathbf{n-1}) \cong \Sigma_{n-1}$ by omitting the n-th slot. We let $\mathscr{D}(\sigma_i) : \mathscr{D}(\mathbf{n}) \to \mathscr{D}(\mathbf{n-1})$ be the composite

(2.9)



To promote the structure further to $\mathscr{D} \in \mathrm{RMod}_{\mathscr{C}}[\Lambda^{op}[\mathscr{V}]]$, we need a series of morphisms

$$\phi^{\lambda}_{k,n_1,\cdots,n_k} : \mathscr{D}(\mathbf{k}) \otimes \mathscr{C}(\mathbf{n_1}) \otimes \cdots \mathscr{C}(\mathbf{n_k}) \to \mathscr{D}(\mathbf{n}) \tag{2.10}$$

where $n \leqslant n_1 + \cdots + n_k$; $\lambda \in \Lambda(\mathbf{n}, \mathbf{n_1} + \cdots + \mathbf{n_k})$. These data restrict to $\psi^{\sigma}_{k,n_1,\cdots,n_k}$ on $\Sigma(\mathbf{n_1} + \cdots + \mathbf{n_k}, \mathbf{n_1} + \cdots + \mathbf{n_k})$ and they must satisfy the following additional requirements:

(1) for $\lambda_1 \in \Lambda(\mathbf{m}, \mathbf{n_1} + \cdots + \mathbf{n_k})$, $\lambda_2 \in \Lambda(\mathbf{l}, \mathbf{m})$, we have

$$\begin{array}{ccc} \mathscr{D}(\mathbf{k}) \otimes \mathscr{C}(\mathbf{n_1}) \otimes \cdots \otimes \mathscr{C}(\mathbf{n_k}) & \xrightarrow{\phi^{\lambda_1}} & \mathscr{D}(\mathbf{m}) \\ & \searrow^{\phi^{\lambda_1\lambda_2}} & \downarrow{\scriptstyle \mathscr{D}(\lambda_2)} \\ & & \mathscr{D}(\mathbf{l}) \end{array}$$

(2) for $\lambda \in \Lambda(\mathbf{m}, \mathbf{n_1} + \cdots + \mathbf{n_{k-1}})$, $\sigma_i \in \Lambda(\mathbf{k}, \mathbf{k-1})$ that omits the i-th target, we have

$$\begin{array}{ccc} \mathscr{D}(\mathbf{k}) \otimes \mathscr{C}(\mathbf{n_1}) \otimes \cdots \otimes \mathscr{I} \otimes \cdots \otimes \mathscr{C}(\mathbf{n_{k-1}}) & \xrightarrow{\eta} & \mathscr{D}(\mathbf{k}) \otimes \mathscr{C}(\mathbf{n_1}) \otimes \cdots \otimes \mathscr{C}(\mathbf{0}) \otimes \cdots \otimes \mathscr{C}(\mathbf{n_{k-1}}) \\ {\scriptstyle \mathscr{D}(\sigma_i)\otimes\cong}\downarrow & & \downarrow{\scriptstyle \phi^{\lambda}_{k,n_1,\cdots,0,\cdots,n_{k-1}}} \\ \mathscr{D}(\mathbf{k-1}) \otimes \mathscr{C}(\mathbf{n_1}) \otimes \cdots \otimes \mathscr{C}(\mathbf{n_{k-1}}) & \xrightarrow[\phi^{\lambda}_{k-1,n_1,\cdots,n_{k-1}}]{} & \mathscr{D}(\mathbf{m}) \end{array}$$

where $\mathscr{I}$ is inserted in the i-th term.

We already have $\phi^{\sigma}_{k,n_1,\cdots,n_k} = \psi^{\sigma}_{k,n_1,\cdots,n_k}$ for $\sigma \in \Sigma_n$, and we use diagram (1) to define $\phi^{\lambda}_{k,n_1,\cdots,n_k}$ for general λ. It remains to show that these data satisfy diagram (2), and because of diagram (1) it suffices to show for $m = n_1 + \cdots + n_{k-1}$ and $\lambda \in \Sigma(\mathbf{m}, \mathbf{n_1} + \cdots + \mathbf{n_{k-1}})$. From (2.9), we have the dotted arrow in the diagram below and the upper triangle commutes. The lower triangle commutes by the associativity of $\mathscr{D}$ as a module over $\mathscr{C}$ in $\Sigma^{op}[\mathscr{V}]$.

$$\begin{array}{ccc} \mathscr{D}(\mathbf{k}) \otimes \mathscr{C}(\mathbf{n_1}) \otimes \cdots \mathscr{C}(\mathbf{n_{k-1}}) & \xrightarrow{\mathrm{id}\otimes\eta\otimes\mathrm{id}} & \mathscr{D}(\mathbf{k}) \otimes \mathscr{C}(\mathbf{n_1}) \otimes \cdots \otimes \mathscr{C}(\mathbf{0}) \otimes \cdots \mathscr{C}(\mathbf{n_{k-1}}) \\ {\scriptstyle \mathscr{D}(\sigma_i)\otimes\mathrm{id}}\downarrow & \swarrow_{\mathscr{D}(\sigma_i)_0} & \downarrow{\scriptstyle \phi^{\lambda}_{k,\ldots}=\psi^{\lambda}_{k,\ldots}} \\ \mathscr{D}(\mathbf{k-1}) \otimes \mathscr{C}(\mathbf{n_1}) \otimes \cdots \mathscr{C}(\mathbf{n_{k-1}}) & \xrightarrow[\phi^{\lambda}_{k-1,\ldots}=\psi^{\lambda}_{k-1,\ldots}]{} & \mathscr{D}(\mathbf{n_1} + \cdots + \mathbf{n_{k-1}}) \end{array}$$

Finally, the diagram (2.9) is functorial with respect to $\mathscr{D} \in \mathrm{RMod}_{\mathscr{C}}[\Sigma^{op}[\mathscr{V}]]$. Therefore, the resulting Λ-sequence structure on $\mathscr{D}$ is functorial and so are diagrams in (1) and (2). As a consequence, we obtain a functorial assignment $r\colon \mathrm{RMod}_{\mathscr{C}}[\Sigma^{op}[\mathscr{V}]] \to \mathrm{RMod}_{\mathscr{C}}[\Lambda^{op}[\mathscr{V}]]$, which gives an inverse for the functor induced by the forgetful map.

Alternatively, one can show that the following diagram is pullback square, and reduce the theorem to the case $\mathscr{C} = \mathscr{I}_1$, which is shown in [Fre09, 5.1.8].

$$\begin{array}{ccc} \mathrm{RMod}_{\mathscr{C}}[\Lambda^{op}[\mathscr{V}]] & \longrightarrow & \mathrm{RMod}_{\mathscr{I}_1}[\Lambda^{op}[\mathscr{V}]] \\ \downarrow & & \downarrow \\ \mathrm{RMod}_{\mathscr{C}}[\Sigma^{op}[\mathscr{V}]] & \longrightarrow & \mathrm{RMod}_{\mathscr{I}_1}[\Sigma^{op}[\mathscr{V}]]. \end{array}$$

Here, the vertical maps are forgetful maps as in Proposition 2.5 and horizontal maps are induced by $\eta : \mathscr{I}_1 \to \mathscr{C}$. □

Corollary 2.11. *For $\mathscr{D} \in \Lambda^{op}[\mathscr{V}]$, the functor $- \odot_\Sigma \mathscr{D} : \Sigma^{op}[\mathscr{V}] \to \Sigma^{op}[\mathscr{V}]$ factors through $\Lambda^{op}[\mathscr{V}]$.*

Proof. We use Theorem 2.6 to identify $\Lambda^{op}[\mathscr{V}]$ with right $\mathscr{I}_1$-modules in $\Sigma^{op}[\mathscr{V}]$, then we use Proposition 2.2. □

For an operad $\mathscr{C}$ in $\mathscr{V}$, there is a notion of a $\mathscr{C}$-algebra in $\mathscr{V}$. Left $\mathscr{C}$-modules generalizes $\mathscr{C}$-algebras in the following way. There is a functor $\iota_0 : \mathscr{V} \to \Lambda^{op}[\mathscr{V}]$ defined by $\iota_0(X)(\mathbf{n}) = \begin{cases} X & n = 0; \\ \emptyset & n \geq 1. \end{cases}$ We also write ι_0 for the functor $\mathscr{V} \to \Sigma^{op}[\mathscr{V}]$ from the same construction that lands in Σ-sequences.

Proposition 2.12. *Let $\mathscr{V}_\mathcal{I}$ denote the category $\mathscr{V}$ under its unit $\mathcal{I}$. For a based operad $\mathscr{C}$ in $\mathscr{V}$, the functor ι_0 restricts to $\mathrm{Alg}_\mathscr{C}[\mathscr{V}_\mathcal{I}] \to \mathrm{LMod}_\mathscr{C}[\Lambda^{op}[\mathscr{V}]_{\mathscr{I}_0}]$; for an operad $\mathscr{C}$ in $\mathscr{V}$, the functor ι_0 restricts to $\mathrm{Alg}_\mathscr{C}[\mathscr{V}] \to \mathrm{LMod}_\mathscr{C}[\Sigma^{op}[\mathscr{V}]]$.*

Proof. For Λ-sequences this is [MZZ25, Lemma 6.5]. The same argument works for Σ-sequences. □

Example 2.13. We list some examples of monoids, box products, and Kelly products.

(1) $\mathscr{I}_0$ is not a monoid as there is no unit map $\mathscr{I}_1 \to \mathscr{I}_0$.
(2) $\mathscr{I}_1^\Sigma$ is a monoid in $\Sigma^{op}[\mathscr{V}]$.

$$\mathrm{LMod}_{\mathscr{I}_1^\Sigma}[\Sigma^{op}[\mathscr{V}]] = \mathrm{RMod}_{\mathscr{I}_1^\Sigma}[\Sigma^{op}[\mathscr{V}]] = \Sigma^{op}[\mathscr{V}]$$

(3) $\mathscr{I}_1$ is a monoid in $\Sigma^{op}[\mathscr{V}]$ and $\Lambda_\mathcal{I}^{op}[\mathscr{V}]$.

$$\begin{aligned} &\mathrm{LMod}_{\mathscr{I}_1}[\Lambda_\mathcal{I}^{op}[\mathscr{V}]] = \mathrm{RMod}_{\mathscr{I}_1}[\Lambda_\mathcal{I}^{op}[\mathscr{V}]] = \Lambda_\mathcal{I}^{op}[\mathscr{V}], \\ &\mathrm{RMod}_{\mathscr{I}_1}[\Sigma^{op}[\mathscr{V}]] = \Lambda^{op}[\mathscr{V}], \\ &\mathrm{LMod}_{\mathscr{I}_1}[\Sigma^{op}[\mathscr{V}]] = \Sigma^{op}[\mathscr{V}]_{\mathscr{C}omm/} \end{aligned}$$

(4) The commutativity operad $\mathscr{C}omm$: $\mathscr{C}omm(\mathbf{n}) = \mathcal{I}$ and the structure maps are all isomorphic to the identity map on $\mathcal{I}$.

$$\begin{aligned} (\mathscr{C}omm \boxtimes \mathscr{C}omm)(\mathbf{n}) &= \oplus_{m=0}^n \mathcal{I}[\Sigma_n/\Sigma_m \times \Sigma_{n-m}] \\ &= \mathcal{I}[\text{partition of } n \text{ into 2 ordered sets}], \\ (\mathscr{C}omm \odot_\Sigma \mathscr{C}omm)(\mathbf{n}) &= \mathcal{I}[\text{partition of } n \text{ into unordered sets}], \\ (\mathscr{C}omm \odot_\Lambda \mathscr{C}omm)(\mathbf{n}) &= \mathcal{I}[\text{partition of } n \text{ into unordered nonempty sets}] \end{aligned}$$

(5) The associativity operad $\mathscr{A}ss$: $\mathscr{A}ss(\mathbf{n}) = \mathcal{I}[\Sigma_n]$ and the structure maps are induced by concatinating and composing permutation of letters.

$$\begin{aligned} \mathscr{A}ss \boxtimes \mathscr{A}ss &= \mathscr{A}ss \\ \mathscr{A}ss \odot_\Sigma \mathscr{A}ss &= \oplus_\mathbb{N} \mathscr{A}ss \\ \mathscr{A}ss \odot_\Lambda \mathscr{A}ss &= \mathscr{A}ss \end{aligned}$$

Proposition 2.14. *Let $\mathscr{C}$ be an operad in a Cartesian monoidal category, then $\mathscr{B} = \mathscr{C} \boxtimes \mathscr{C}omm$ is also an operad. When $\mathscr{C}$ is based, $\mathscr{B}$ is also based.*

Proof. In a Cartesian monoidal category, $\mathscr{C}omm$ is the terminal object in both $\Sigma^{op}[\mathscr{V}]$ and $\Lambda^{op}[\mathscr{V}]$. The multiplication map of $\mathscr{B}$ is given by $(\odot = \odot_\Sigma)$

$$\begin{aligned}
\mathscr{B}\odot\mathscr{B} &\cong (\mathscr{C}\odot(\mathscr{C}\boxtimes\mathscr{C}omm))\boxtimes(\mathscr{C}omm\odot(\mathscr{C}\boxtimes\mathscr{C}omm)) && \text{by Theorem 2.1(1)}\\
&\to (\mathscr{C}\odot\mathscr{C})\boxtimes(\mathscr{C}\odot\mathscr{C}omm)\boxtimes(\mathscr{C}omm\odot(\mathscr{C}\boxtimes\mathscr{C}omm)) && \text{by Theorem 2.1(5)}\\
&\to (\mathscr{C}\odot\mathscr{C})\boxtimes\mathscr{C}omm && \mathscr{C}omm \text{ is terminal}\\
&\to \mathscr{C}\boxtimes\mathscr{C}omm = \mathscr{B}. && \text{structure map of } \mathscr{C}
\end{aligned}$$

The unit map $\eta : \mathscr{I}_1^\Sigma \cong \mathscr{I}_1^\Sigma\boxtimes\mathscr{I}_0 \to \mathscr{C}\boxtimes\mathscr{C}omm$ is the $\boxtimes$-product of the unit map $\mathscr{I}_1^\Sigma\to\mathscr{C}$ and the unique map $\mathscr{I}_0\to\mathscr{C}omm$. This is associative and unital. When $\mathscr{C}$ is based, just replace $\odot_\Sigma$ by $\odot_\Lambda$ and $\mathscr{I}_1^\Sigma$ by $\mathscr{I}_1$. □

Remark 2.15. More explicitly,

$$\begin{aligned}
\mathscr{B}(0) &= \mathscr{C}(0),\\
\mathscr{B}(1) &= \mathscr{C}(0)\sqcup\mathscr{C}(1),\\
\text{and } \mathscr{B}(n) &= \sqcup_{S\subset\{1,2,\cdots,n\}}\mathscr{C}(|S|).
\end{aligned}$$

Take any $S\subset\{1,\cdots,k\}$ and $T_i\subset\{1,\cdots,n_i\}$ for $1\le i\le k$. Let $T\subset\{1,\cdots,\Sigma_{i=1}^k n_i\}$ be the union $\cup_{i\in S}(T_i+\Sigma_{j<i,j\in S}|T_j|)$, where T_i+n is $\{x+n|x\in T_i\}$ for a number n. There are the multiplication maps

$$\mathscr{C}(|S|)\times\times_{i=1}^k\mathscr{C}(|T_i|)\xrightarrow[\text{if } i\notin S]{\text{project to }*}\mathscr{C}(|S|)\times\times_{i\in S}\mathscr{C}(|T_i|)\xrightarrow{\mu_\mathscr{C}}\mathscr{C}(|T|),$$

coproducts of which give the multiplication map

$$\mu_\mathscr{B} : \mathscr{B}(k)\times\times_{i=1}^k\mathscr{B}(n_i)\to\mathscr{B}(\Sigma_{i=1}^k n_i).$$

Remark 2.16. The map

$$\mathscr{C}\cong\mathscr{C}\boxtimes\mathscr{I}_0\xrightarrow{\mathrm{id}\boxtimes\eta_+}\mathscr{C}\boxtimes\mathscr{C}omm=\mathscr{B} \tag{2.17}$$

is a map of operads. Restriction along it endows $\mathscr{B}$-algebras the structure of $\mathscr{C}$-algebras. In contrast, a $\mathscr{C}$-algebra X can be made into a $\mathscr{B}$-algebra via the same maneuver as in Remark 2.15. Namely, the structure maps $\theta_\mathscr{B} : \mathscr{B}(n)\times X^n\to X$ are coproducts of

$$\mathscr{C}(|S|)\times\times_{i=1}^n X\xrightarrow[\text{if } i\notin S]{\text{project to }*}\mathscr{C}(|S|)\times\times_{i\in S}X\xrightarrow{\theta_\mathscr{C}}X$$

for $S\subset\{1,2,\cdots,n\}$. This embeds the category of $\mathscr{C}$-algebras as a fully faithful subcategory of $\mathscr{B}$-algebras.

Although $\mathscr{I}_0$ is not an operad in our sense, it still makes sense to consider $\mathscr{C}=\mathscr{I}_0$ as a heuristic example. The category of $\mathscr{I}_0$-algebras are based objects. It embeds into the category of commutative algebras as the trivial ones, where the commutative algebra structure maps are all in the form of $X^n\to *\to X$.

Remark 2.18. There are maps of operads $\mathscr{C}omm\to\mathscr{B}\to\mathscr{C}omm$. Here, the first map is

$$\mathscr{C}omm\cong\mathscr{I}_0\boxtimes\mathscr{C}omm\xrightarrow{\eta_+\boxtimes\mathrm{id}}\mathscr{C}\boxtimes\mathscr{C}omm=\mathscr{B},$$

or explicitely $*\to\mathscr{C}(0)\to\mathscr{B}(n)$ at each level, and the second map uses that $\mathscr{C}omm$ is terminal. This embeds commutative algebras as a faithful subcategory of $\mathscr{B}$-algebras.

2.3. **Relation to category of operators.** Let $\mathscr{F}$ denote the category with objects $\mathbf{n}$ and based maps; let $\mathscr{F}_{>0} \subset \mathscr{F}$ denote the category with objects $\mathbf{n}$ and only morphisms $\phi : \mathbf{m} \to \mathbf{n}$ such that $\phi^{-1}(0) = \{0\}$ (May–Thomason [MT78] call these effective morphisms and Lurie [Lur09] call these active morphisms.) Denote Π the subcategory of $\mathscr{F}$ with morphisms $\phi : \mathbf{m} \to \mathbf{n}$ such that each $\phi^{-1}(j)$ has at most one element for every $1 \leqslant j \leqslant n$. Note that $\mathscr{F}$ is the skeleton of the category of finite based sets, $\mathscr{F}_{>0}$ is the skeleton of the category of finite sets and $\Pi \cap \mathscr{F}_{>0}$ is exactly Λ.

Recall that $\mathscr{V}$ is a complete and cocomplete closed monoidal category in this section. When category of operators is involved, we assume further that $\mathscr{V}$ is Cartesian monoidal. A category is considered as a $\mathscr{V}$-enriched category by tensoring the morphism sets with $\mathcal{I}$.

Definition 2.19. ([May09, Section 1]) A $\mathscr{V}$-category of operators $\mathscr{D}$ is a category enriched over $\mathscr{V}$ with objects the same as $\mathscr{F}$, such that the inclusion $\Pi \to \mathscr{F}$ factors through as an inclusion $\Pi \to \mathscr{D}$ and a surjection $\mathscr{D} \to \mathscr{F}$.

Definition 2.20. For an operad $\mathscr{C}$ in a Cartesian monoidal category $\mathscr{V}$, there is an associated $\mathscr{V}$-enriched symmetric monoidal category $\widehat{\mathscr{C}}$ with morphisms

$$\widehat{\mathscr{C}}(\mathbf{n}, \mathbf{m}) = \coprod_{\phi \in \mathscr{F}(\mathbf{n}, \mathbf{m})} \mathscr{C}(\phi^{-1}(1)) \otimes \cdots \otimes \mathscr{C}(\phi^{-1}(m)).$$

Identity maps are given by $\mathrm{id} \in \mathscr{F}(\mathbf{n}, \mathbf{n})$ and each unit $\mathcal{I} \to \mathscr{C}(\mathbf{1})$, and the composition is induced from the unit $\mathcal{I} \to \mathscr{C}(\mathbf{1})$, projection maps $\mathscr{C}(\mathbf{n}) \to *$ and the operad structure maps γ.

Example 2.21. We have $\Pi = \widehat{\mathscr{I}_1}$, $\mathscr{F} = \widehat{\mathscr{C}omm}$.

When $\mathscr{C}$ is a unital operad in a Cartesian monoidal category $\mathscr{V}$ with $\mathscr{C}(\mathbf{n}) \neq \varnothing$ for all $\mathbf{n}$, $\widehat{\mathscr{C}}$ is a category of operators where the map $\Pi \to \widehat{\mathscr{C}}$ is induced by $\mathscr{I}_1 \to \mathscr{C}$ and the map $\widehat{\mathscr{C}} \to \mathscr{F}$ is induced by $\mathscr{C} \to \mathscr{C}omm$.

Definition 2.22. [MZZ25, Section 12] Let $\mathscr{C}$ be an operad in $\mathscr{V}$. The permutative envelope of $\mathscr{C}$ is the category $\overline{\mathscr{C}}$ with objects $\mathbf{n}$ and morphisms given by

$$\overline{\mathscr{C}}(\mathbf{m}, \mathbf{n}) = \mathscr{C}^{\boxtimes n}(\mathbf{m}).$$

See [MZZ25] for compositions in $\overline{\mathscr{C}}$. Explicitly, there is

$$\overline{\mathscr{C}}(\mathbf{n}, \mathbf{m}) = \coprod_{\phi \in \mathscr{F}_{>0}(\mathbf{n}, \mathbf{m})} \mathscr{C}(\phi^{-1}(1)) \otimes \cdots \otimes \mathscr{C}(\phi^{-1}(m))$$

and $\overline{\mathscr{C}}$ is a ($\mathscr{V}$-enriched symmetric monoidal) subcategory of $\widehat{\mathscr{C}}$.

Example 2.23. We have $\Lambda = \overline{\mathscr{I}_1}$, $\mathscr{F}_{>0} = \overline{\mathscr{C}omm}$.

Proposition 2.24. *For any based operad $\mathscr{C}$, we have isomorphisms of Λ^{op}-sequences*

$$\overline{\mathscr{C}}(-, \mathbf{n}) \cong \mathscr{C}^{\boxtimes n}; \tag{2.25}$$

$$\widehat{\mathscr{C}}(-, \mathbf{n}) \cong \overline{\mathscr{C}}(-, \mathbf{n}) \boxtimes \mathscr{C}omm \cong \mathscr{C}^{\boxtimes n} \boxtimes \mathscr{C}omm. \tag{2.26}$$

Proof. We have $\overline{\mathscr{C}}(-,\mathbf{n}) = \mathscr{C}^{\boxtimes n}$ from [MZZ25, Lemma 12.13]. For $\widehat{\mathscr{C}}$, we have

$$\begin{aligned}
\overline{\mathscr{C}}(\mathbf{m},\mathbf{n}) \boxtimes \mathscr{C}omm &= \int^{\mathbf{m}_1,\mathbf{m}_2} \Sigma(\mathbf{m},\mathbf{m}_1+\mathbf{m}_2) \otimes \overline{\mathscr{C}}(\mathbf{m}_1,\mathbf{n}) \otimes \mathscr{C}omm(\mathbf{m}_2) \\
&= \int^{\mathbf{m}_1,\mathbf{m}_2} \Sigma(\mathbf{m},\mathbf{m}_1+\mathbf{m}_2) \otimes \coprod_{\phi\in\mathscr{F}_{>0}(\mathbf{m}_1,\mathbf{n})} \mathscr{C}(\phi^{-1}(1)) \otimes \cdots \otimes \mathscr{C}(\phi^{-1}(n)) \\
&\cong \coprod_{\{\text{partition of } \mathbf{m}=S\vee T\}} \Big(\coprod_{\phi\in\mathscr{F}_{>0}(|S|,\mathbf{n})} \mathscr{C}(\phi^{-1}(1)) \otimes \cdots \otimes \mathscr{C}(\phi^{-1}(n)) \Big) \\
&\cong \coprod_{\phi\in\mathscr{F}(\mathbf{m},\mathbf{n})} \mathscr{C}(\phi^{-1}(1)) \otimes \cdots \otimes \mathscr{C}(\phi^{-1}(n)) \\
&= \widehat{\mathscr{C}}(\mathbf{m},\mathbf{n}).
\end{aligned}$$

□

One can start with (2.26) as the definition of $\widehat{\mathscr{C}}$ and use the same arguments as in [MZZ25, Section 12.1] to show that $\widehat{\mathscr{C}}$ has the structure of a symmetric monoidal category.

Remark 2.27. Taking $\mathscr{B} = \mathscr{C} \boxtimes \mathscr{C}omm$ as in Proposition 2.14, there are maps of $\mathscr{V}$-enriched symmetric monoidal categories $\overline{\mathscr{C}} \to \overline{\mathscr{B}} \to \widehat{\mathscr{C}} \to \widehat{\mathscr{B}}$. The first and the last maps are induced by the map of operads (2.17); the middle map can be constructed explicitly on morphism spaces, or in perspective of (2.25) via the maps $\mathscr{C}omm^{\boxtimes n} \to \mathscr{C}omm$ to the terminal operad. For this reason, we speculate that the operad $\mathscr{B}$ has an interesting role in the infinite loop space theory, which is unexplored.

2.4. **Abundance or scarcity of monads.** We review a general monad construction in [May09] and show that applied to the permutative envolop $\overline{\mathscr{C}}$ and its subcategories Σ and Λ, the monad is isomorphic to the monad associated to the monoid $\mathscr{C}$. We identify the category of right $\mathscr{C}$-modules completely.

Definition 2.28. Let $\mathscr{D}$ be a category enriched over $\mathscr{V}$, and Ξ be a subcategory of $\mathscr{D}$. We say Ξ is a freely-enriched wide subcategory of $\mathscr{D}$ if it is an ordinary category consisting of all objects of $\mathscr{D}$ and viewed as a subcategory of $\mathscr{D}$ via morphisms $i : \coprod_{\Xi(X,Y)} \mathcal{I} \to \mathscr{D}(X,Y)$ for objects $X, Y \in \mathscr{D}$.

Let $\mathscr{D}$ be a category enriched over $\mathscr{V}$ and Ξ be a freely-enriched wide subcategory of $\mathscr{D}$. Recall that for functors $X : \Xi \to \mathscr{V}$ and $Y : \Xi^{op} \to \mathscr{V}$, $Y \otimes_\Xi X$ is the coend [1]

$$\int^{\mathbf{m}\in\Xi} Y(\mathbf{m}) \otimes X(\mathbf{m}).$$

Definition 2.29. ([May09, Construction 2.1]) The associated monad $\mathbb{D}_\Xi$ in $\Xi[\mathscr{V}]$ is given by

$$\mathbb{D}_\Xi X(\mathbf{n}) = \mathscr{D}(-,\mathbf{n}) \otimes_\Xi X.$$

One can check that the definition indeed give monads. For example, the unit map $\eta : X \to \mathbb{D}_\Xi X$ is given by

$$X(\mathbf{n}) = \int^{\mathbf{m}} \Xi(\mathbf{m},\mathbf{n}) \otimes X(\mathbf{m}) \xrightarrow{i_*} \int^{\mathbf{m}} \mathscr{D}(\mathbf{m},\mathbf{n}) \otimes X(\mathbf{m}) = \mathbb{D}_\Xi X(n).$$

The product map is induced by compositions in $\mathscr{D}$.

We may also define a monad on contravariant functors.

[1] We write objects of Ξ as $\mathbf{m}$ because of our applications. Here we used that Ξ is an ordinary category.

Definition 2.30. The associated monad $\mathbb{D}_{\Xi}^{op}$ in $\Xi^{op}[\mathscr{V}]$ is given by

$$\mathbb{D}_{\Xi}^{op}Y(\mathbf{n}) = Y \otimes_{\Xi} \mathscr{D}(\mathbf{n}, -).$$

We prove a categorical fact identifying the algebras, a special case of which is stated in [May09, Proposition 2.5] without proof.

Proposition 2.31. *Suppose that Ξ is a freely-enriched wide subcategory of a $\mathscr{V}$-enriched category $\mathscr{D}$. There are isomorphisms of categories:*

$$\mathrm{Fun}_{\mathscr{V}}(\mathscr{D}, \mathscr{V}) \cong \mathrm{Alg}_{\mathbb{D}_{\Xi}}[\Xi[\mathscr{V}]]$$
$$\mathrm{Fun}_{\mathscr{V}}(\mathscr{D}^{op}, \mathscr{V}) \cong \mathrm{Alg}_{\mathbb{D}_{\Xi}^{op}}[\Xi^{op}[\mathscr{V}]]$$

Proof. We prove the first case and the second case is obtained by taking $\mathscr{D}$ to be $\mathscr{D}^{op}$ and Ξ to be Ξ^{op} in the first case. We write $\iota : \Xi \to \mathscr{D}$ for the inclusion of the subcategory. The monad $\mathbb{D}_{\Xi}$ is indeed the monad of the free-forgetful adjunction of $\iota^* : \mathrm{Fun}_{\mathscr{V}}(\mathscr{D}, \mathscr{V}) \to \mathrm{Fun}_{\mathscr{V}}(\Xi, \mathscr{V})$, and we show that this is a monadic adjunction by Beck's monadicity theorem and then we are done. As Ξ is a wide subcategory, ι^* reflects isomorphisms. Now we take functors $F, G \in \mathrm{Fun}_{\mathscr{V}}(\mathscr{D}, \mathscr{V})$, $H \in \mathrm{Fun}_{\mathscr{V}}(\Xi, \mathscr{V})$ and a split coequalizer diagram

$$\iota^*F \underset{\iota^*g}{\overset{\iota^*f}{\rightrightarrows}} \iota^*G \underset{s}{\overset{e}{\rightleftarrows}} H.$$

We define $K \in \mathrm{Fun}_{\mathscr{V}}(\mathscr{D}, \mathscr{V})$ by setting $K(\mathbf{m}) = H(\mathbf{m})$ on objects and $K = e \circ G$ on morphisms as in the commutative diagram

$$\begin{array}{ccc} \Xi(\mathbf{m},\mathbf{n}) & \xrightarrow{H} & \underline{\mathrm{Hom}}(H(\mathbf{m}), H(\mathbf{n})) \\ {\scriptstyle\iota}\downarrow & \overset{K}{\nearrow} & \uparrow{\scriptstyle e} \\ \mathscr{D}(\mathbf{m},\mathbf{n}) & \xrightarrow[G]{} & \underline{\mathrm{Hom}}(G(\mathbf{m}), G(\mathbf{n})) \end{array}$$

Then K is a coequalizer of f, g and $\iota^*K = H$. □

Let $\mathscr{C}$ be a based operad in $\mathscr{V}$. We apply Definition 2.30 to the permutative envelope $\overline{\mathscr{C}}$ and the category of operators $\widehat{\mathscr{C}}$ and get four associated monads $\overline{\mathbb{C}}_{\Sigma}^{op}$, $\overline{\mathbb{C}}_{\Lambda}^{op}$, $\widehat{\mathbb{C}}_{\Sigma}^{op}$, $\widehat{\mathbb{C}}_{\Lambda}^{op}$. Proposition 2.31 identifies algebras over these monads in the corresponding categories.

Lemma 2.32. *The monads $\overline{\mathbb{C}}_{\Sigma}^{op}$ and $\overline{\mathbb{C}}_{\Lambda}^{op}$ are isomorphic to the monads $- \odot \mathscr{C}$ associated to the monoid $\mathscr{C}$ as right endofunctors on $\Sigma^{op}[\mathscr{V}]$ and $\Lambda^{op}[\mathscr{V}]$.*

Proof. We show for Λ and the case for Σ is similar. By definitions, we have

$$(\overline{\mathbb{C}}_{\Lambda}^{op}Y)(\mathbf{n}) = \int^{\mathbf{m}} Y(\mathbf{m}) \otimes \overline{\mathscr{C}}(\mathbf{n}, \mathbf{m}) = \int^{\mathbf{m}} Y(\mathbf{m}) \otimes \mathscr{C}^{\boxtimes m}(\mathbf{n}) = (Y \odot_{\Lambda} \mathscr{C})(\mathbf{n}).$$

The structure maps $\mu : \overline{\mathbb{C}}_{\Lambda}^{op}\overline{\mathbb{C}}_{\Lambda}^{op}Y \to \overline{\mathbb{C}}_{\Lambda}^{op}Y$ and $\eta : \overline{C}_{\Lambda}^{op}Y \to Y$ are exactly the right module structure maps $\mu' : Y \odot_{\Lambda} \mathscr{C} \odot_{\Lambda} \mathscr{C} \to Y \odot_{\Lambda} \mathscr{C}$ and $\eta' : Y \odot_{\Lambda} \mathscr{C} \to Y$. □

Theorem 2.33. *The following categories are isomorphic within each group.*

(1) *(a)* $\mathrm{Fun}_{\mathscr{V}}(\overline{\mathscr{C}}^{op}, \mathscr{V})$.
(b) $\mathrm{Alg}_{\overline{\mathbb{C}}_{\Sigma}^{op}}[\Sigma^{op}[\mathscr{V}]]$
(c) $\mathrm{Alg}_{\overline{\mathbb{C}}_{\Lambda}^{op}}[\Lambda^{op}[\mathscr{V}]]$
(d) $\mathrm{RMod}_{\mathscr{C}}[\Sigma^{op}[\mathscr{V}]]$

(e) $\mathrm{RMod}_{\mathscr{C}}[\Lambda^{op}[\mathscr{V}]]$

(2) (a) $\mathrm{Fun}_{\mathscr{V}}(\widehat{\mathscr{C}^{op}}, \mathscr{V})$.
(b) $\mathrm{Alg}_{\widehat{\mathbb{C}}^{op}_{\Sigma}}[\Sigma^{op}[\mathscr{V}]]$
(c) $\mathrm{Alg}_{\widehat{\mathbb{C}}^{op}_{\Lambda}}[\Lambda^{op}[\mathscr{V}]]$

In parituclar, this gives an alternative proof to Theorem 2.6.

Proof. Proposition 2.31 shows the isomorphisms of (1a) (1b) (1c) and of (2a) (2b) (2c). Lemma 2.32 shows that (1c) ≅ (1e) and (1b) ≅ (1d). □

Example 2.34. We have:

(1) The category of right modules over $\mathscr{I}_1$ in $\Sigma^{op}[\mathscr{V}]$ or in $\Lambda^{op}[\mathscr{V}]$ is isomorphic to the category of contravariant functors from Λ to $\mathscr{V}$.
(2) The category of right modules over $\mathscr{C}omm$ in $\Sigma^{op}[\mathscr{V}]$ or in $\Lambda^{op}[\mathscr{V}]$ is isomorphic to the category of contravariant functors from $\mathscr{F}_{>0}$ to $\mathscr{V}$.

3. Monoidal localization

3.1. **Localization.** Let $\mathscr{V}$ be a category and S be a family of morphisms in $\mathscr{V}$. Gabriel and Zisman [GZ12] constructed a category $\mathscr{V}[S^{-1}]$ called the localization of $\mathscr{V}$ with respect to S and a functor $L : \mathscr{V} \to \mathscr{V}[S^{-1}]$ with the following properties:

(1) $L(s)$ is an isomorphism for any morphism s in S.
(2) For any functor $F : \mathscr{V} \to \mathscr{W}$ such that for any s in S the image $F(s)$ is an isomorphism, there exists a unique functor $\tilde{F} : \mathscr{V}[S^{-1}] \to \mathscr{W}$ such that $F = \tilde{F} \circ L$.

The localization $\mathscr{V}[S^{-1}]$ can be characterized as follows. The objects of $\mathscr{V}[S^{-1}]$ are the same as those of $\mathscr{V}$. For morphisms, we consider the "necklaces" in $\mathscr{V}$, that is, diagrams in $\mathscr{V}$ of the form

$$\leftarrow \cdots \leftarrow\rightarrow \cdots \rightarrow\leftarrow \cdots \leftarrow\rightarrow \cdots \rightarrow$$

where each $\rightarrow$ represents a morphism in $\mathrm{Mor}(\mathscr{V})$ and each $\leftarrow$ represents a morphism in S. For $A, B \in \mathrm{Ob}(\mathscr{V}[S^{-1}]) = \mathrm{Ob}(\mathscr{V})$, we take $\mathrm{Map}_{\mathscr{V}[S^{-1}]}(A, B)$ to be the set of all necklaces in $\mathscr{V}$ that connect A and B quotient the following relationships.

(1) $\xrightarrow{f}\xrightarrow{g}\sim\xrightarrow{g\circ f}$.
(2) For $s : A \to B$ in S, $\xrightarrow{s}\xleftarrow{s}\sim id_A$, $\xleftarrow{s}\xrightarrow{s}\sim id_B$.

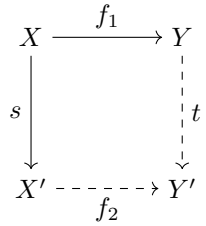
The composition is taken in the obvious manner. We take $L : \mathscr{V} \to \mathscr{V}[S^{-1}]$ as the embedding. It can be checked that the above universal property is satisfied.

Our aim is to localize a symmetric monoidal category $\mathscr{V}$ so that tensor products commute with small or finite colimits. A straightforward attempt is to simply take S as the collection of all morphisms of the form $\{\mathrm{colim}_{i\in I}(A_i \otimes X) \to \mathrm{colim}_{i\in I}(A_i) \otimes X\}$, where I runs over all small categories and X runs over objects of $\mathscr{V}$. To ensure that $\mathscr{V}[S^{-1}]$ have enough colimits, Gabriel and Zisman introduced the following property of S.

Definition 3.1. [GZ12, Definition 2.2.] Let $\mathscr{V}$ be a category and S be a collection of morphisms. The collection S is called a *right multiplicative system* (or S admits a calculus of left fractions) if S satisfies the following properties.

(1) The identity morphisms of $\mathscr{V}$ are all in S.
(2) If $f : A \to B$, $g : B \to C$ are two morphisms in S, then $g \circ f : A \to C$ is in S.

(3) Suppose we are given two morphisms $f_1 : X \to Y$ and $s : X \to X'$ with s belonging to S, then there are two morphisms $f_2 : X' \to Y'$ and $t : Y \to Y'$ where t belongs to S and the following diagram commutes



(4) Suppose we are given two morphisms $g_1, g_2 : X \rightrightarrows Y$ in $\mathscr{V}$ and a morphism $s : W \to X$ in S such that $g_1 \circ s = g_2 \circ s$, then there is an object Z and a morphism $t : Y \to Z$ in S such that $t \circ g_1 = t \circ g_2$.

We define *left multiplicative systems* (or a collection of morphisms that admits a calculus of right fractions) dually. We say S is a *multiplicative system* if it is both a right and a left multiplicative system.

Definition 3.2. Fix a cardinal κ, a multiplicative system $S \subset \mathrm{Mor}(\mathscr{V})$ is called *κ-closed* if it is closed under κ-small products and coproducts of morphisms; it is called *closed* if it is closed under small products and coproducts of morphisms.

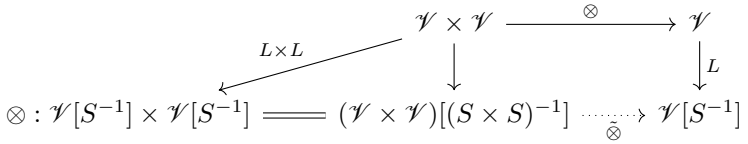
Proposition 3.3. *Let S be a multiplicative system in $\mathscr{V}$.*

- *The localization $L : \mathscr{V} \to \mathscr{V}[S^{-1}]$ preserves finite limits and colimits that exist in $\mathscr{V}$ (*[GZ67, Prop 3.1]*).*
- *If S is κ-closed, the localization $L : \mathscr{V} \to \mathscr{V}[S^{-1}]$ preserves κ-small limits and colimits that exist in $\mathscr{V}$ (*[Kra10, Prop 3.5.1]*).*

Definition 3.4. A collection of morphisms $S \subset \mathrm{Mor}(\mathscr{V})$ is called *monoidal* if it is closed under tensoring with identity maps $\mathrm{id}_X : X \to X$ for all $X \in \mathscr{V}$ ([Day73, page 1]).

Proposition 3.5. *Let S be monoidal. Then the localization $L : \mathscr{V} \to \mathscr{V}[S^{-1}]$ is strong symmetric monoidal.*

Proof. This is due to [Day73, Corollary 1.4] and we present an alternative proof. We note that the localization $L : \mathscr{V} \to \mathscr{V}[S^{-1}]$ induces a functor $L \times L : \mathscr{V} \times \mathscr{V} \to \mathscr{V}[S^{-1}] \times \mathscr{V}[S^{-1}]$. We take $S \times S \subset \mathrm{Mor}(\mathscr{V} \times \mathscr{V})$ and formally we have $(\mathscr{V} \times \mathscr{V})[(S \times S)^{-1}] \cong \mathscr{V}[S^{-1}] \times \mathscr{V}[S^{-1}]$. Since we have stipulated that S is closed under finite composition and tensor with identities, we have that for $(s_1, s_2) \in S \times S$, $L(s_1 \otimes s_2) = L(\mathrm{id} \otimes s_2) \circ L(s_1 \otimes \mathrm{id})$ is invertible in $\mathscr{V}[S^{-1}]$. Therefore, by the universal property there exists a unique functor $\tilde{\otimes} : \mathscr{V}[S^{-1}] \times \mathscr{V}[S^{-1}] \to \mathscr{V}[S^{-1}]$ such that $\tilde{\otimes} \circ (L \times L) = L \circ \otimes$. This endows $\mathscr{V}[S^{-1}]$ with an inherited symmetric monoidal structure.



□

We also denote the induced tensor product on $\mathscr{V}[S^{-1}]$ by $\otimes$. From the discussions above, we have $LA \otimes LB = L(A \otimes B)$.

3.2. **Monoidal multiplicative closure.**

Construction 3.6. For $S \subset \mathrm{Mor}(\mathscr{V})$ containing all identity morphisms, we define a collection of morphisms S_α for each ordinal α by transfinite induction.

We set $S_0 = S$. For an ordinal α, suppose that S_β has been constructed for all $\beta < \alpha$. When α is a limit ordinal, we define $S_\alpha = \cup_{\beta<\alpha} S_\beta$. When $\alpha = \beta + 1$, we define S_α as follows.

(1) Let $T_{\alpha,0} = S_\beta$. Let $T_{\alpha,1}$ be the collection of all morphisms of the form $s_1 : Y \to Y \coprod_X Z$, where $s_0 : X \to Z$ is in $T_{\alpha,0}$ and $f_1 : X \to Y$ is in $\mathscr{V}$.

$$\begin{array}{ccc} X & \xrightarrow{f_1} & Y \\ {\scriptstyle s_0}\downarrow & & \vdots\,{\scriptstyle s_1} \\ Z & \dashrightarrow & Y \coprod_X Z \end{array}$$

Note that $T_{\alpha,1}$ contains $T_{\alpha,0}$.

(2) $T_{\alpha,2}$ consists of morphisms in $T_{\alpha,1}$ and all canonical morphisms

$$s_2 : \mathrm{coeq}(f \circ s_1, g \circ s_1) \to \mathrm{coeq}(f, g),$$

where $f, g : X \rightrightarrows Y$ are two morphisms in $\mathscr{V}$ and $s_1 : W \to X$ is in $T_{\alpha,1}$.

$$W \xrightarrow{s_1} X \underset{g}{\overset{f}{\rightrightarrows}} Y \xrightarrow{\mathrm{can}} \mathrm{coeq}(f \circ s_1, g \circ s_1), \quad \mathrm{can} \searrow \vdots\, s_2 \ \text{to} \ \mathrm{coeq}(f, g)$$

(3) $T_{\alpha,3}$ is the collection of all morphisms of the form $s_3 : Y \prod_X Z \to Y$, where $s_2 : Z \to X$ is in $T_{\alpha,2}$ and $f_1 : Y \to X$ is in $\mathscr{V}$.

$$\begin{array}{ccc} Y \prod_X Z & \dashrightarrow & Z \\ {\scriptstyle s_3}\downarrow & & \downarrow{\scriptstyle s_2} \\ Y & \xrightarrow[f_1]{} & X \end{array}$$

Note $T_{\alpha,3}$ contains $T_{\alpha,2}$.

(4) $T_{\alpha,4}$ consists of morphisms in $T_{\alpha,3}$ and all canonical morphisms

$$s_4 : \mathrm{eq}(f, g) \to \mathrm{eq}(s_3 \circ f, s_3 \circ g),$$

where $f, g : X \rightrightarrows Y$ are two morphisms in $\mathscr{V}$ and $s_3 : Y \to Z$ is in $T_{\alpha,3}$.

$$\begin{array}{ccccccc} \mathrm{eq}(f,g) & & & & & & \\ {\scriptstyle s_4}\vdots & \searrow^{\mathrm{can}} & & & & & \\ \mathrm{eq}(s_3 \circ f, s_3 \circ g) & \xrightarrow{\mathrm{can}} & X & \underset{g}{\overset{f}{\rightrightarrows}} & Y & \xrightarrow{s_3} & Z \end{array}$$

(5) $T_{\alpha,5}$ consists of all finite compositions of morphisms in $T_{\alpha,4}$.

(6) $T_{\alpha,6}$ consists of all α-small coproducts and products of morphisms in $T_{\alpha,5}$.

(7) $T_{\alpha,7}$ consists of all morphisms of the form $\mathrm{id}_X \otimes s_6$ or $s_6 \otimes \mathrm{id}_X$, where s_6 is in $T_{\alpha,6}$ and X is an object in $\mathscr{V}$.

We set $S_\alpha := T_{\alpha,7}$.

Theorem 3.7. *For a regular cardinal κ, the $\overline{S} := S_\kappa$ in Construction 3.6 is a κ-closed monoidal multiplicative system containing S. It has the universal property that if $\overline{S}'$ is a κ-closed monoidal multiplicative system containing S, then the localization $L' : \mathscr{V} \to \mathscr{V}[(\overline{S}')^{-1}]$ factors through $L : \mathscr{V} \to \mathscr{V}[\overline{S}^{-1}]$ uniquely.*

We refer to $\overline{S}$ as the κ-closed monoidal multiplicative closure of S. We also have $\cup_\kappa S_\kappa$ as the closed monoidal multiplicative closure of S, and it shares a similar universal property.

Proof. By construction, we have $S = S_0 \subset S_1 \subset \cdots \subset S_\kappa = \bigcup_{\beta<\kappa} S_\beta = \overline{S}$.

(1) We first show that $\overline{S}$ is both a right and a left multiplicative system, i.e. $\overline{S}$ satisfies Definition 3.1 items (1) to (4) and the dual statements. It is obvious from the construction that $\overline{S}$ contains all identities and is closed under finite composition. For any morphism $f_1 : X \to Y$ in $\mathscr{V}$ and $s : X \to X'$ in $S_\beta \subset \overline{S}$, the induced map $s' : Y \to Y \coprod_X X'$ is in $S_{\beta+1} \subset S_\kappa$. Suppose that we are given two morphisms $g_1, g_2 : X \rightrightarrows Y$ in $\mathscr{V}$ and a morphism $s : W \to X$ in $S_\beta \subset \overline{S}$ such that $g_1 \circ s = g_2 \circ s$. We may take $Z = \mathrm{coeq}(f, g)$ and $t : Y \cong \mathrm{coeq}(f \circ s, g \circ s) \to \mathrm{coeq}(f, g) = Z$. Then t belongs to $S_{\beta+1} \subset \overline{S}$. Dually, the corresponding assertions for left multiplicative systems also hold.

(2) We then check $\overline{S}$ is closed under κ-small product and coproduct. For any ordinal $\alpha < \kappa$ and $s_1, \dots s_\alpha \in \overline{S}$, for each $i \leq \alpha$ there exists an ordinal $f(i) < \kappa$ such that $s_i \in S_{f(i)}$. Take $\beta_0 = \mathrm{Sup}\{f(1), \cdots, f(\alpha)\}$. As κ is a regular cardinal, we must have $\beta_0 < \kappa$. By construction we have $\coprod_i s_i \in S_{\beta_0+1} \subset S_\kappa$ and $\prod_i s_i \in S_{\beta_0+1} \subset S_\kappa$. It is easy to see that $\overline{S}$ is closed under tensoring with identities.

(3) It remains to show the universal property. Suppose we are given another κ-closed monoidal multiplicative system containing S, denoted by $\overline{S}'$. We use induction to show that for any ordinal $\alpha \leq \kappa$, L' takes morphisms in S_α to isomorphisms in $\mathscr{V}[\overline{S}'^{-1}]$. The base case is true because $S_0 = S$. Suppose that $L'(S_\beta)$ are isomorphisms for all $\beta < \alpha$, and we would like to show it for $L'(S_\alpha)$. If α is a limit ordinal, the claim is true by the definition of S_α. If $\alpha = \beta + 1$ for some β, using the fact that L' preserves κ-small limits, colimits and the tensor product (Proposition 3.3 and Proposition 3.5), one can show that L' sends each class in Construction 3.6 to isormorphisms. For example, L' sends $T_{\alpha,1}$ to isomorphisms because it does for $T_{\alpha,0}$ and the pushout of an isomorphism is also an isomorphism. The assertion will then follow from the universal property of localization. □

3.3. **Closed monoidal localization.**

Lemma 3.8. *Let J be a small diagram and $\mathcal{B}$ be a collection of morphisms in J. Let $J' = J[\mathcal{B}^{-1}]$ be the diagram obtained by adding an inverse to each $f \in \mathcal{B}$. Then the inclusion $\iota : J \to J'$ is cofinal* [2].

Proof. We need to show that the comma category $x \downarrow \iota$ is non-empty and connected for all objects $x \in J'$. Obviously, it contains the object $(x \in J, \mathrm{id} : x \to x \in J')$. Suppose we are given any object $(y \in J,\ F : x \to y \in J')$. Then F may be written as a finite composite

$$x = x_0 \xrightarrow{F_1} x_1 \xrightarrow{F_2} x_2 \xrightarrow{F_3} \cdots \xrightarrow{F_n} x_n = y,$$

[2]We use cofinal to mean final in [ML98, IX.3], as it concerns colimits.

where each F_i is either in J or is the formal inverse of some morphism in $\mathcal{B}$. Write $F_0 = \mathrm{id}_x$ and $G_i = F_i \circ F_{i-1} \circ \cdots \circ F_0$. We show that there is a zigzag of morphisms in $x \downarrow \iota$ from (x, G_0) to (x, G_n) by providing morphisms between pairs (x, G_{i-1}) and (x, G_i).

- Case 1: $F_i = g$ for $g \in J$. The diagram $\begin{array}{ccc} x & = & x \\ {\scriptstyle G_{i-1}}\downarrow & & \downarrow{\scriptstyle G_i} \\ x_{i-1} & \xrightarrow[F_i]{} & x_i \end{array}$ is a morphism from (x, G_{i-1}) to (x, G_i).
- Case 2: $F_i = f^{-1}$ for $f \in \mathcal{B}$. The diagram $\begin{array}{ccc} x & = & x \\ {\scriptstyle G_{i-1}}\downarrow & & \downarrow{\scriptstyle G_i} \\ x_{i-1} & \xleftarrow[f]{} & x_i \end{array}$ is a morphism from (x, G_i) to (x, G_{i-1}). □

Theorem 3.9. *Let $S = \{\mathrm{colim}_{i\in I}(A_i \otimes X) \to \mathrm{colim}_{i\in I}(A_i) \otimes X\}$ where I runs over all κ-small categories and X runs over all objects of $\mathscr{V}$. Let $\overline{S}$ be its κ-closed monoidal multiplicative closure. Then*

(1) The localization $L : \mathscr{V} \to L\mathscr{V} := \mathscr{V}[\overline{S}^{-1}]$ preserves κ-small limits, colimits and the tensor product.

(2) For any κ-small diagram $(A_j)_{j\in J}$ in $\mathscr{V}$ and object $X \in \mathscr{V}$, the induced map

$$\mathrm{colim}_{j\in J}(LA_j \otimes LX) \to \mathrm{colim}_{j\in J}(LA_j) \otimes LX$$

is an isomorphism in $L\mathscr{V}$.

(3) For any κ-small diagram $(B_i)_{i\in I}$ in $L\mathscr{V}$ and object $X \in L\mathscr{V}$, the natural map

$$\mathrm{colim}_{i\in I}(B_i \otimes X) \to (\mathrm{colim}_{i\in I} B_i) \otimes X$$

is an isomorphism in $L\mathscr{V}$.

Proof. Item (1) follows from Proposition 3.3 and Proposition 3.5. Item (2) follows from (1).

It remains to show item (3). We use X to denote its lift in $\mathscr{V}$. Using necklaces, we can lift each morphism of diagram $D_0 = (B_i)_{i\in I}$ in $L\mathscr{V}$ to a necklaces in $\mathscr{V}$, resulting in a κ-small diagram $D = (A_j)_{j\in J}$ in $\mathscr{V}$, where $\mathrm{ob}(I) \subset \mathrm{ob}(J)$ and $A_i = B_i$ for $i \in I$.

$$\mathrm{r} \xleftarrow{s} \mathrm{j} \xrightarrow{f} \mathrm{t}$$
$$\mathrm{J}$$

$$B_r \xleftarrow{D(s)} A_j \xrightarrow{D(f)} B_t$$
$$D = (A_j) \text{ in } \mathscr{V}$$

$$\mathrm{r} \longrightarrow \mathrm{t} \qquad \mathrm{r} \xrightarrow{s} \mathrm{j} \xrightarrow{f} \mathrm{t}$$
$$\mathrm{I} \qquad\qquad \mathrm{K}$$

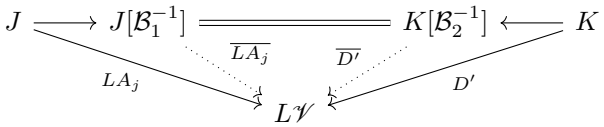
$$B_r \longrightarrow B_t \qquad B_r \xrightarrow[\sim]{D'(s)} A_j \xrightarrow{LD(f)} B_t$$
$$D_0 = (B_i) \text{ in } L\mathscr{V} \qquad\qquad D' \text{ in } L\mathscr{V}$$

By construction, if we reversed all backward arrows in the necklaces, we obtained another κ-small category K with the same objects as J. As the backward arrows in J are sent to isomorphisms in $L\mathscr{V}$, we can construct a diagram $D' : K \to L\mathscr{V}$ such that $D' = LD$ on objects and forward arrows in the necklaces, $D'(s) = D(s)^{-1}$ if s is a backward arrow in the necklaces J, and extending to other morphisms by composition. Then I is a subcategory of K and $(B_i)_{i\in I} = (D')|_I$. It may be shown that $I \subset K$ is a cofinal subcategory, and so

$$\operatorname{colim}_K(D') \cong \operatorname{colim}_I(B_i) \tag{3.10}$$

$$\operatorname{colim}_K(D' \otimes X) \cong \operatorname{colim}_I(B_i \otimes X) \tag{3.11}$$

Denote the class of all the backward arrows in J by $\mathcal{B}_1$ and the corresponding reversed arrows in K by $\mathcal{B}_2$. Then $J[\mathcal{B}_1^{-1}] \cong K[\mathcal{B}_2^{-1}]$, and the dotted arrows exist and are equal by universal properties:



There is

$$\operatorname{colim}_J(LA_j) \cong \operatorname{colim}(\overline{LA_j}) \cong \operatorname{colim}(\overline{D'}) \cong \operatorname{colim}_K(D'), \tag{3.12}$$

where the first and last isomorphisms are by Lemma 3.8. Similarly, there is

$$\operatorname{colim}_J(LA_j \otimes X) \cong \operatorname{colim}_K(D' \otimes X). \tag{3.13}$$

Note that in this case we used that if s is a backward arrow in the necklaces, then both $LD(s) \otimes \mathrm{id}_X$ and $D'(x) \otimes \mathrm{id}_X$ are isomorphisms in $L\mathscr{V}$, which is guaranteed by the monoidal property of $\overline{S}$.

Combing (3.12), (3.13) and item (2), we have

$$\operatorname{colim}_K(D' \otimes X) \cong (\operatorname{colim}_K D') \otimes X.$$

With (3.10) and (3.11), this finishes the proof. □

4. Normal oplax monoidal structure

Let $(\mathscr{V}, \otimes, \mathcal{I})$ be a symmetric monoidal category and L be the monoidal localization functor defined in Theorem 3.9. In this section we discuss the monoidal products of Λ-sequences when $\otimes$ in $\mathscr{V}$ is not assumed to commute with colimits.

4.1. **n-fold Day convolution.** For a finite set S and an object X in $\mathscr{V}$, define $S \otimes X = \amalg_S X$ to be the S-fold coproduct of the object X. There is

$$S \otimes (T \otimes X) \cong (S \times T) \otimes X \text{ for finite sets } S, T \text{ and } X \in \mathscr{V}.$$

Proposition 4.1. *The functor $- \otimes X : \mathrm{FinSet} \to \mathscr{V}$ commutes with finite colimits; and the functor $T \otimes - : \mathscr{V} \to \mathscr{V}$ commutes with colimits.*

Proof. It follows from the definition. □

Warning 4.2. For $X, Y \in \mathscr{V}$, $S \otimes (X \otimes Y)$ can be identified with $(S \otimes X) \otimes Y$ only when $\otimes$ commutes with finite colimits in $\mathscr{V}$. We shall be careful where to put the parenthesis.

Definition 4.3. For $\mathscr{D}_1, \dots, \mathscr{D}_n \in \Lambda^{op}[\mathscr{V}]$, the n-fold box product is defined to be

$$\mathscr{D}_1 \boxtimes_\Lambda \cdots \boxtimes_\Lambda \mathscr{D}_n = \int^{i_1,\dots,i_n} \Lambda(-, \mathbf{i_1} + \cdots + \mathbf{i_n}) \otimes \Big(\mathscr{D}_1(\mathbf{i_1}) \otimes \cdots \otimes \mathscr{D}_n(\mathbf{i_n}) \Big)$$

Remark 4.4. There are maps $\mathscr{D}_1 \boxtimes_\Lambda \mathscr{D}_2 \boxtimes_\Lambda \mathscr{D}_3 \to (\mathscr{D}_1 \boxtimes_\Lambda \mathscr{D}_2) \boxtimes_\Lambda \mathscr{D}_3$ and $\mathscr{D}_1 \boxtimes_\Lambda \mathscr{D}_2 \boxtimes_\Lambda \mathscr{D}_3 \to \mathscr{D}_1 \boxtimes_\Lambda (\mathscr{D}_2 \boxtimes_\Lambda \mathscr{D}_3)$ by the universal property. These are only isomorphisms when $\otimes$ commutes with colimits. In fact, $(\Lambda^{op}[\mathscr{V}], \boxtimes_\Lambda, \mathscr{I}_0)$ forms a normal op-lax monoidal structure as defined later in Definition 4.17, but we shall omit the details.

Proposition 4.5. *For $\mathscr{D}_1, \dots, \mathscr{D}_n \in \Lambda^{op}[\mathscr{V}]$, the natural map*

$$\iota : \mathscr{D}_1 \boxtimes_\Sigma \cdots \boxtimes_\Sigma \mathscr{D}_n \to \mathscr{D}_1 \boxtimes_\Lambda \cdots \boxtimes_\Lambda \mathscr{D}_n$$

is an isomorphism of Σ-sequences.

Proof. As left Kan extension can be computed pointwise by colimits over the arrow category, it suffices to prove that in the diagram

$$\begin{array}{ccc} \Sigma^{op} \times \cdots \times \Sigma^{op} & \xrightarrow{i \times \cdots \times i} & \Lambda^{op} \times \cdots \times \Lambda^{op} \\ {\scriptstyle \vee_\Sigma}\downarrow & & \downarrow{\scriptstyle \vee_\Lambda} \\ \Sigma^{op} & \xrightarrow[i]{} & \Lambda^{op} \end{array}$$

the induced map of arrow categories $i : \vee_\Sigma \downarrow \mathbf{n} \to \vee_\Lambda \downarrow \mathbf{n}$ is cofinal for every $\mathbf{n} \in \Sigma$. The exact same proof of [MZZ25, Theorem 3.4] works. □

4.2. **2-fold Kelly product.** Recall that when $(\mathscr{V}, \otimes, \mathcal{I})$ is closed symmetric monoidal, the Kelly product is defined to be the coend

$$\begin{aligned} \mathscr{D}_1 \odot_\Lambda \mathscr{D}_2 &:= \int^{\mathbf{k}} \mathscr{D}_1(\mathbf{k}) \otimes \mathscr{D}_2^{\boxtimes k} \\ &\cong \int^{\mathbf{k}, \mathbf{n_1}, \dots, \mathbf{n_k}} \Lambda(-, \mathbf{n_1} + \cdots + \mathbf{n_k}) \otimes \mathscr{D}_1(\mathbf{k}) \otimes \mathscr{D}_2(\mathbf{n_1}) \otimes \cdots \otimes \mathscr{D}_2(\mathbf{n_k}) \end{aligned}$$

These equivalent definitions involve several isomorphisms commuting $\otimes$ with colimits. Inspired by it, for $\mathscr{D}_1, \mathscr{D}_2 \in \Lambda^{op}[\mathscr{V}]_{\mathscr{I}_0}$, we set

$$\begin{aligned} \mathscr{D}_1 \otimes \mathscr{D}_2^{\otimes k} : \Lambda^{op} \times (\Lambda^{op})^k &\to \mathscr{V} \\ (\mathbf{m}, \mathbf{n_1}, \cdots, \mathbf{n_k}) &\mapsto \mathscr{D}_1(\mathbf{m}) \otimes \mathscr{D}_2(\mathbf{n_1}) \otimes \cdots \otimes \mathscr{D}_2(\mathbf{n_k}). \end{aligned}$$

Let $(\mathscr{D}_1 \wedge \mathscr{D}_2)_{\mathbf{k}} : \Lambda^{op} \times \Lambda^{op} \to \mathscr{V}$ be the left Kan extension as in:

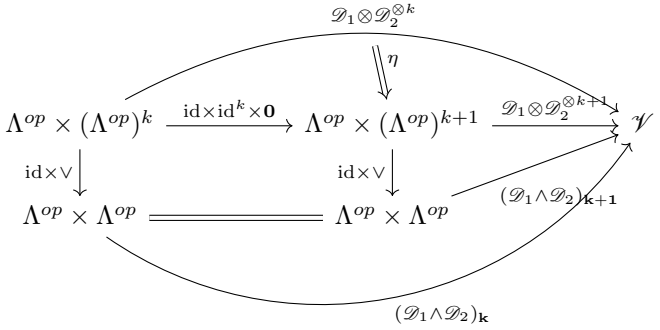
$$\begin{array}{ccc} \Lambda^{op} \times (\Lambda^{op})^k & \xrightarrow{\mathscr{D}_1 \otimes \mathscr{D}_2^{\otimes k}} & \mathscr{V} \\ {\scriptstyle \mathrm{id} \times \vee}\downarrow & \nearrow_{(\mathscr{D}_1 \wedge \mathscr{D}_2)_{\mathbf{k}}} & \\ \Lambda^{op} \times \Lambda^{op} & & \end{array}$$

Explicitly,

$$(\mathscr{D}_1 \wedge \mathscr{D}_2)_{\mathbf{k}}(\mathbf{m}, -) = \int^{\mathbf{n_1}, \cdots, \mathbf{n_k}} \Lambda(-, \mathbf{n_1} + \cdots + \mathbf{n_k}) \otimes \big(\mathscr{D}_1(\mathbf{m}) \otimes \mathscr{D}_2(\mathbf{n_1}) \otimes \cdots \otimes \mathscr{D}_2(\mathbf{n_k}) \big).$$

The functors $(\mathscr{D}_1 \wedge \mathscr{D}_2)_{\mathbf{k}}(-,-)$ are also functorial for $\mathbf{k}$ in Λ. To see this, the non-trivial check is for the morphisms of the form $\sigma_{k+1} : \mathbf{k} \to \mathbf{k} + \mathbf{1}$. We examine the

following diagrams of definition for $(\mathscr{D}_1 \wedge \mathscr{D}_2)_{\mathbf{k}}$ and $(\mathscr{D}_1 \wedge \mathscr{D}_2)_{\mathbf{k+1}}$.



The natural transformation η is induced by $\mathcal{I} \to \mathscr{D}_2(\mathbf{0})$, and we have natural transformations

$$(\mathscr{D}_1 \wedge \mathscr{D}_2)_{\mathbf{k}} \to \mathrm{LKan}_{\mathrm{id} \times \vee} \mathscr{D}_1 \otimes \mathscr{D}_2^{\otimes k} \otimes \mathscr{D}_2(\mathbf{0}) \to (\mathscr{D}_1 \wedge \mathscr{D}_2)_{\mathbf{k+1}},$$

where the first arrow is induced by η and the second arrow is induced by $\mathrm{id} \times \mathrm{id}^k \times \mathbf{0}$.

Definition 4.6. For $\mathscr{D}_1, \mathscr{D}_2 \in \Lambda^{op}[\mathscr{V}]_{\mathscr{I}_0}$, define the 2-fold Kelly product to be

$$\begin{aligned} \mathscr{D}_1 \odot_\Lambda \mathscr{D}_2 &= \int^{\mathbf{k}} (\mathscr{D}_1 \wedge \mathscr{D}_2)_{\mathbf{k}}(\mathbf{k}, -) \\ &= \int^{\mathbf{k}, \mathbf{n_1}, \ldots, \mathbf{n_k}} \Lambda(-, \mathbf{n_1} + \cdots + \mathbf{n_k}) \otimes \big(\mathscr{D}_1(\mathbf{k}) \otimes \mathscr{D}_2(\mathbf{n_1}) \otimes \cdots \otimes \mathscr{D}_2(\mathbf{n_k})\big) \end{aligned}$$

Proposition 4.7. *Composing with $L : \mathscr{V} \to L\mathscr{V}$ gives functors*

$$L : \Lambda^{op}[\mathscr{V}] \to \Lambda^{op}[L\mathscr{V}] \text{ and } L : \Lambda^{op}[\mathscr{V}]_{\mathscr{I}_0} \to \Lambda^{op}[L\mathscr{V}]_{\mathscr{I}_0}$$

that preserve $\boxtimes_\Lambda$ and $\odot_\Lambda$.

Proof. This is because L preserves colimits and tensor products in $\mathscr{V}$. □

We denote $\mathscr{I}_1 = \Lambda(-, \mathbf{1}) \otimes \mathcal{I}$. It is a unital Λ-sequence.

Example 4.8. There are two canonical morphisms $\alpha_{1,0,0} : \mathscr{D} \to \mathscr{I}_1 \odot_\Lambda \mathscr{D}$ and $\alpha_{1,1,0} : \mathscr{D} \to \mathscr{D} \odot_\Lambda \mathscr{I}_1$, which we define now. By the Yoneda Lemma, there is

$$\begin{aligned} \mathscr{D} &= \int^{\mathbf{n_1}} \Lambda(-, \mathbf{n_1}) \otimes \mathscr{D}(\mathbf{n_1}) \\ &= \int^{\mathbf{k}, \mathbf{n_1}, \cdots, \mathbf{n_k}} (\Lambda(-, \mathbf{n_1} + \cdots + \mathbf{n_k}) \times \Lambda(\mathbf{k}, \mathbf{1})) \otimes \mathscr{D}(\mathbf{n_1}) \otimes \cdots \otimes \mathscr{D}(\mathbf{n_k}) \end{aligned}$$

By commuting tensor products with coproducts as in

$$\Lambda(\mathbf{k}, \mathbf{1}) \otimes (\mathcal{I} \otimes X) \to (\Lambda(\mathbf{k}, \mathbf{1}) \otimes \mathcal{I}) \otimes X,$$

the map $\alpha_{1,0,0}$ maps $\mathscr{D}$ to

$$\mathscr{I}_1 \odot_\Lambda \mathscr{D} = \int^{\mathbf{k}, \mathbf{n_1}, \cdots, \mathbf{n_k}} \Lambda(-, \mathbf{n_1} + \cdots + \mathbf{n_k}) \otimes \left((\Lambda(\mathbf{k}, \mathbf{1}) \otimes \mathcal{I}) \otimes \mathscr{D}(\mathbf{n_1}) \otimes \cdots \otimes \mathscr{D}(\mathbf{n_k})\right).$$

And similarly for $\alpha_{1,1,0}$.

Remark 4.9. When $\otimes$ commutes with colimits in $\mathscr{V}$, both $\alpha_{1,0,0}$ and $\alpha_{1,1,0}$ are isomorphisms. In general, $\mathscr{I}_1$ is not necessarily isomorphic to $\mathscr{I}_1 \odot_\Lambda \mathscr{I}_1$.

4.3. **3-fold Kelly product.** Now we consider the 3-fold product. Since we do not have associativity any more, we have to make separate definitions.

Let

$$\mathscr{D}_1 \otimes \mathscr{D}_2^{\otimes k} \otimes \mathscr{D}_3^{\otimes t} : \Lambda^{op} \times (\Lambda^{op})^k \times (\Lambda^{op})^t \to \mathscr{V}$$

$(\mathbf{m}, \mathbf{n_1}, \cdots, \mathbf{n_k}, \mathbf{j_1}, \cdots, \mathbf{j_t}) \mapsto \mathscr{D}_1(\mathbf{m}) \otimes \mathscr{D}_2(\mathbf{n_1}) \otimes \cdots \otimes \mathscr{D}_2(\mathbf{n_k}) \otimes \mathscr{D}_3(\mathbf{j_1}) \otimes \cdots \otimes \mathscr{D}_3(\mathbf{j_t})$, and $(\mathscr{D}_1 \wedge \mathscr{D}_2 \wedge \mathscr{D}_3)_{\mathbf{k},\mathbf{t}} : \Lambda^{op} \times \Lambda^{op} \times \Lambda^{op} \to \mathscr{V}$ be the left Kan extension as in:

$$\begin{array}{ccc} \Lambda^{op} \times (\Lambda^{op})^k \times (\Lambda^{op})^t & \xrightarrow{\mathscr{D}_1 \otimes \mathscr{D}_2^{\otimes k} \otimes \mathscr{D}_3^{\otimes t}} & \mathscr{V} \\ {\scriptstyle \mathrm{id} \times \vee \times \vee}\downarrow & \nearrow_{(\mathscr{D}_1 \wedge \mathscr{D}_2 \wedge \mathscr{D}_3)_{\mathbf{k},\mathbf{t}}} & \\ \Lambda^{op} \times \Lambda^{op} \times \Lambda^{op} & & \end{array}$$

Similarly as before, the functors $(\mathscr{D}_1 \wedge \mathscr{D}_2 \wedge \mathscr{D}_3)_{\mathbf{k},\mathbf{t}}$ are also functorial for $\mathbf{k}, \mathbf{t}$ in Λ.

Definition 4.10. For $\mathscr{D}_1, \mathscr{D}_2, \mathscr{D}_3 \in \Lambda^{op}[\mathscr{V}]_{\mathscr{I}_0}$, define the 3-fold Kelly product to be

$$\begin{aligned} \mu_3(\mathscr{D}_1, \mathscr{D}_2, \mathscr{D}_3) &= \int^{\mathbf{k},\mathbf{t}} (\mathscr{D}_1 \wedge \mathscr{D}_2 \wedge \mathscr{D}_3)_{\mathbf{k},\mathbf{t}}(\mathbf{k}, \mathbf{t}, -) \\ &= \int^{\substack{\mathbf{k},\mathbf{t} \\ \mathbf{n_1},\cdots,\mathbf{n_k} \\ \mathbf{j_1},\cdots,\mathbf{j_t}}} \big(\Lambda(-, \mathbf{j_1} + \cdots + \mathbf{j_t}) \times \Lambda(\mathbf{t}, \mathbf{n_1} + \cdots + \mathbf{n_k})\big) \\ &\quad \otimes \Big(\mathscr{D}_1(\mathbf{k}) \otimes \mathscr{D}_2(\mathbf{n_1}) \otimes \cdots \otimes \mathscr{D}_2(\mathbf{n_k}) \otimes \mathscr{D}_3(\mathbf{j_1}) \otimes \cdots \otimes \mathscr{D}_3(\mathbf{j_t})\Big) \end{aligned}$$

Lemma 4.11. *We have the following isomorphisms:*

- *For any object $X \in \mathscr{V}$ which may be dependent on*

$$\mathbf{k}, \mathbf{n_1}, \mathbf{j}_{1,1}, \cdots, \mathbf{j}_{1,\mathbf{n_1}}, \mathbf{n_k}, \mathbf{j}_{\mathbf{k},1}, \cdots, \mathbf{j}_{\mathbf{k},\mathbf{n_k}},$$

there is

$$\begin{aligned} &\int^{\mathbf{r_1},\cdots,\mathbf{r_k}} \big(\Lambda(-, \mathbf{r_1} + \cdots + \mathbf{r_k}) \\ &\qquad \times \Lambda(\mathbf{r_1}, \mathbf{j}_{1,1} + \cdots + \mathbf{j}_{1,\mathbf{n_1}}) \times \cdots \times \Lambda(\mathbf{r_k}, \mathbf{j}_{\mathbf{k},1} + \cdots + \mathbf{j}_{\mathbf{k},\mathbf{n_k}})\big) \otimes X \\ &\cong \Lambda(-, \mathbf{j}_{1,1} + \cdots + \mathbf{j}_{1,\mathbf{n_1}} + \cdots + \mathbf{j}_{\mathbf{k},1} + \cdots + \mathbf{j}_{\mathbf{k},\mathbf{n_k}}) \otimes X; \end{aligned} \tag{4.12}$$

- *For any object $X \in \mathscr{V}$ which may be dependent on $\mathbf{k}, \mathbf{n_1}, \cdots, \mathbf{n_k}$, there is*

$$\int^{\mathbf{t},\mathbf{j_1},\cdots,\mathbf{j_t}} \big(\Lambda(-, \mathbf{j_1} + \cdots + \mathbf{j_t}) \times \Lambda(\mathbf{t}, \mathbf{n_1} + \cdots + \mathbf{n_k})\big) \otimes \big(X \otimes \mathscr{D}_3(\mathbf{j_1}) \otimes \cdots \otimes \mathscr{D}_3(\mathbf{j_t})\big) \tag{4.13}$$

$$\cong \int^{\mathbf{j_1},\cdots,\mathbf{j}_{\mathbf{n_1}+\cdots+\mathbf{n_k}}} \Lambda(-, \mathbf{j_1} + \cdots + \mathbf{j}_{\mathbf{n_1}+\cdots+\mathbf{n_k}}) \otimes \big(X \otimes \mathscr{D}_3(\mathbf{j_1}) \otimes \cdots \otimes \mathscr{D}_3(\mathbf{j}_{\mathbf{n_1}+\cdots+\mathbf{n_k}})\big)$$

Proof. For (4.12), observe that

$$\begin{aligned} &\int^{\mathbf{r_1},\cdots,\mathbf{r_k}} \big(\Lambda(-, \mathbf{r_1} + \cdots + \mathbf{r_k}) \\ &\qquad \times \Lambda(\mathbf{r_1}, \mathbf{j}_{1,1} + \cdots + \mathbf{j}_{1,\mathbf{n_1}}) \times \cdots \times \Lambda(\mathbf{r_k}, \mathbf{j}_{\mathbf{k},1} + \cdots + \mathbf{j}_{\mathbf{k},\mathbf{n_k}})\big) \otimes X \\ \cong &\int^{\mathbf{r_k}} \cdots \int^{\mathbf{r_1}} \big(\Lambda(-, \mathbf{r_1} + \cdots + \mathbf{r_k}) \\ &\qquad \times \Lambda(\mathbf{r_1}, \mathbf{j}_{1,1} + \cdots + \mathbf{j}_{1,\mathbf{n_1}}) \times \cdots \times \Lambda(\mathbf{r_k}, \mathbf{j}_{\mathbf{k},1} + \cdots + \mathbf{j}_{\mathbf{k},\mathbf{n_k}})\big) \otimes X \end{aligned}$$

By the Yoneda lemma, we have

$$\Lambda(-,\mathbf{j_{1,1}}+\cdots+\mathbf{j_{1,n_1}}+\mathbf{r_2}+\cdots+\mathbf{r_k}) \cong \int^{\mathbf{r_1}} \Lambda(-,\mathbf{r_1}+\cdots+\mathbf{r_k})\times\Lambda(\mathbf{r_1},\mathbf{j_{1,1}}+\cdots+\mathbf{j_{1,n_1}})$$

The claim follows by repeating the process for $\mathbf{r_2},\cdots,\mathbf{r_k}$ and Proposition 4.1. For (4.13), let $F:\Lambda^{op}\to\mathscr{V}$ be given by

$$F(\mathbf{t}) = \int^{\mathbf{j_1},\cdots,\mathbf{j_t}} \Lambda(-,\mathbf{j_1}+\cdots+\mathbf{j_t})\otimes(X\otimes\mathscr{D}_3(\mathbf{j_1})\otimes\cdots\otimes\mathscr{D}_3(\mathbf{j_t})).$$

By the Yoneda lemma, $\int^{\mathbf{t}}\Lambda(\mathbf{t},\mathbf{n_1}+\cdots+\mathbf{n_k})\otimes F(\mathbf{t}) \cong F(\mathbf{n_1}+\cdots+\mathbf{n_k})$. The claim follows from Proposition 4.1. □

Proposition 4.14. *There are two natural transformations*

$$s_1:\mu_3(\mathscr{D}_1,\mathscr{D}_2,\mathscr{D}_3)\to(\mathscr{D}_1\odot_\Lambda\mathscr{D}_2)\odot_\Lambda\mathscr{D}_3 \text{ and } s_2:\mu_3(\mathscr{D}_1,\mathscr{D}_2,\mathscr{D}_3)\to\mathscr{D}_1\odot_\Lambda(\mathscr{D}_2\odot_\Lambda\mathscr{D}_3).$$

When $\otimes$ commutes with colimits in $\mathscr{V}$, s_1 and s_2 are isomorphisms.

Proof. We have

$$\begin{aligned}
&(\mathscr{D}_1\odot_\Lambda\mathscr{D}_2)\odot_\Lambda\mathscr{D}_3\\
&=\int^{\mathbf{t},\mathbf{j_1},\cdots,\mathbf{j_t}}\Lambda(-,\mathbf{j_1}+\cdots+\mathbf{j_t})\otimes\Big((\mathscr{D}_1\odot_\Lambda\mathscr{D}_2)(\mathbf{t})\otimes\mathscr{D}_3(\mathbf{j_1})\otimes\cdots\otimes\mathscr{D}_3(\mathbf{j_t})\Big)\\
&=\int^{\mathbf{t},\mathbf{j_1},\cdots,\mathbf{j_t}}\Lambda(-,\mathbf{j_1}+\cdots+\mathbf{j_t})\otimes\Big(\Big(\int^{\mathbf{k},\mathbf{n_1},\cdots,\mathbf{n_k}}\Lambda(\mathbf{t},\mathbf{n_1}+\cdots+\mathbf{n_k})\\
&\quad\otimes\mathscr{D}_1(\mathbf{k})\otimes\mathscr{D}_2(\mathbf{n_1})\otimes\cdots\otimes\mathscr{D}_2(\mathbf{n_k})\Big)\otimes\mathscr{D}_3(\mathbf{j_1})\otimes\cdots\otimes\mathscr{D}_3(\mathbf{j_t})\Big).
\end{aligned}$$

Comparing with the formula in Definition 4.10, the natural transformation s_1 is given by the natural map from coend of tensors to the tensor with the coend. Notice that s_1 is pointwise in $\overline{S}$.

$$\begin{aligned}
&\mathscr{D}_1\odot_\Lambda(\mathscr{D}_2\odot_\Lambda\mathscr{D}_3)\\
&=\int^{\mathbf{k},\mathbf{r_1},\cdots,\mathbf{r_k}}\Lambda(-,\mathbf{r_1}+\cdots+\mathbf{r_k})\otimes\big(\mathscr{D}_1(\mathbf{k})\otimes(\mathscr{D}_2\odot_\Lambda\mathscr{D}_3)(\mathbf{r_1})\otimes\cdots\otimes(\mathscr{D}_2\odot_\Lambda\mathscr{D}_3)(\mathbf{r_k})\big)\\
&=\int^{\mathbf{k},\mathbf{r_1},\cdots,\mathbf{r_k}}\Lambda(-,\mathbf{r_1}+\cdots+\mathbf{r_k})\otimes\Big(\mathscr{D}_1(\mathbf{k})\otimes\\
&\quad\int^{\mathbf{n_1},\mathbf{j_{1,1}},\cdots,\mathbf{j_{1,n_1}}}\Lambda(\mathbf{r_1},\mathbf{j_{1,1}}+\cdots+\mathbf{j_{1,n_1}})\otimes(\mathscr{D}_2(\mathbf{n_1})\otimes\mathscr{D}_3(\mathbf{j_{1,1}})\otimes\cdots\otimes\mathscr{D}_3(\mathbf{j_{1,n_1}}))\otimes\cdots\\
&\quad\otimes\int^{\mathbf{n_k},\mathbf{j_{k,1}},\cdots,\mathbf{j_{k,n_k}}}\Lambda(\mathbf{r_k},\mathbf{j_{k,1}}+\cdots+\mathbf{j_{k,n_k}})\otimes(\mathscr{D}_2(\mathbf{n_k})\otimes\mathscr{D}_3(\mathbf{j_{k,1}})\otimes\cdots\otimes\mathscr{D}_3(\mathbf{j_{k,n_k}}))\Big).
\end{aligned}$$

We now define the natural transformation s_2 as the following composite:

$$\begin{aligned}
\mu_3(\mathscr{D}_1,\mathscr{D}_2,\mathscr{D}_3) = &\int^{\substack{\mathbf{k},\mathbf{t}\\ \mathbf{n_1},\cdots,\mathbf{n_k}\\ \mathbf{j_1},\cdots,\mathbf{j_t}}} \left(\Lambda(-,\mathbf{j_1}+\cdots+\mathbf{j_t})\times\Lambda(\mathbf{t},\mathbf{n_1}+\cdots+\mathbf{n_k})\right)\\
&\otimes\left(\mathscr{D}_1(\mathbf{k})\otimes\mathscr{D}_2(\mathbf{n_1})\otimes\cdots\otimes\mathscr{D}_2(\mathbf{n_k})\otimes\mathscr{D}_3(\mathbf{j_1})\otimes\cdots\otimes\mathscr{D}_3(\mathbf{j_t})\right)\\
\text{by (4.13)}\cong&\int^{\substack{\mathbf{k}\\ \mathbf{n_1},\cdots,\mathbf{n_k}\\ \mathbf{j_1},\cdots,\mathbf{j}_{\mathbf{n_1}+\cdots+\mathbf{n_k}}}} \Lambda(-,\mathbf{j_1}+\cdots+\mathbf{j}_{\mathbf{n_1}+\cdots+\mathbf{n_k}})\\
&\otimes\left(\mathscr{D}_1(\mathbf{k})\otimes\mathscr{D}_2(\mathbf{n_1})\otimes\cdots\otimes\mathscr{D}_2(\mathbf{n_k})\otimes\mathscr{D}_3(\mathbf{j_1})\otimes\cdots\otimes\mathscr{D}_3(\mathbf{j}_{\mathbf{n_1}+\cdots+\mathbf{n_k}})\right)\\
\text{by (4.12)}\cong&\int^{\substack{\mathbf{k},\mathbf{r_1},\cdots,\mathbf{r_k}\\ \mathbf{n_1},\mathbf{j}_{1,1},\cdots,\mathbf{j}_{1,\mathbf{n_1}}\\ \cdots\\ \mathbf{n_k},\mathbf{j}_{\mathbf{k},1},\cdots,\mathbf{j}_{\mathbf{k},\mathbf{n_k}}}} \left(\Lambda(-,\mathbf{r_1}+\cdots+\mathbf{r_k})\right.\\
&\times\Lambda(\mathbf{r_1},\mathbf{j}_{1,1}+\cdots+\mathbf{j}_{1,\mathbf{n_1}})\times\cdots\times\left.\Lambda(\mathbf{r_k},\mathbf{j}_{\mathbf{k},1}+\cdots+\mathbf{j}_{\mathbf{k},\mathbf{n_k}})\right)\\
&\otimes\left(\mathscr{D}_1(\mathbf{k})\otimes\mathscr{D}_2(\mathbf{n_1})\otimes\cdots\otimes\mathscr{D}_2(\mathbf{n_k})\otimes\mathscr{D}_3(\mathbf{j}_{1,1})\otimes\cdots\otimes\mathscr{D}_3(\mathbf{j}_{1,\mathbf{n_1}}))\otimes\cdots\right.\\
&\left.\otimes\mathscr{D}_3(\mathbf{j}_{\mathbf{k},1})\otimes\cdots\otimes\mathscr{D}_3(\mathbf{j}_{\mathbf{k},\mathbf{n_k}})\right)\\
\longrightarrow&\int^{\substack{\mathbf{k},\mathbf{r_1},\cdots,\mathbf{r_k}\\ \mathbf{n_1},\mathbf{j}_{1,1},\cdots,\mathbf{j}_{1,\mathbf{n_1}}\\ \cdots\\ \mathbf{n_k},\mathbf{j}_{\mathbf{k},1},\cdots,\mathbf{j}_{\mathbf{k},\mathbf{n_k}}}} \Lambda(-,\mathbf{r_1}+\cdots+\mathbf{r_k})\otimes\left(\mathscr{D}_1(\mathbf{k})\otimes\right.\\
&\left(\Lambda(\mathbf{r_1},\mathbf{j}_{1,1}+\cdots+\mathbf{j}_{1,\mathbf{n_1}})\otimes(\mathscr{D}_2(\mathbf{n_1})\otimes\mathscr{D}_3(\mathbf{j}_{1,1})\otimes\cdots\otimes\mathscr{D}_3(\mathbf{j}_{1,\mathbf{n_1}}))\right)\otimes\cdots\\
&\left.\otimes\left(\Lambda(\mathbf{r_k},\mathbf{j}_{\mathbf{k},1}+\cdots+\mathbf{j}_{\mathbf{k},\mathbf{n_k}})\otimes(\mathscr{D}_2(\mathbf{n_k})\otimes\mathscr{D}_3(\mathbf{j}_{\mathbf{k},1})\otimes\cdots\otimes\mathscr{D}_3(\mathbf{j}_{\mathbf{k},\mathbf{n_k}}))\right)\right)\\
\longrightarrow&\mathscr{D}_1\odot_\Lambda(\mathscr{D}_2\odot_\Lambda\mathscr{D}_3),
\end{aligned}$$

where the first arrow is from coproduct of tensors to tensor with coproducts and the second arrow is from coend of tensors to the tensor with the coend. Notice that s_2 is also pointwise in $\overline{S}$. □

Example 4.15. Writing $\mu_2(\mathscr{D}_1,\mathscr{D}_2)=\mathscr{D}_1\odot_\Lambda\mathscr{D}_2$, there are canonical maps

$$\begin{aligned}
\alpha_{2,0,0}:&\mu_2(\mathscr{D}_1,\mathscr{D}_2)\to\mu_3(\mathscr{I}_1,\mathscr{D}_1,\mathscr{D}_2),\\
\alpha_{2,1,0}:&\mu_2(\mathscr{D}_1,\mathscr{D}_2)\to\mu_3(\mathscr{D}_1,\mathscr{I}_1,\mathscr{D}_2),\\
\alpha_{2,2,0}:&\mu_2(\mathscr{D}_1,\mathscr{D}_2)\to\mu_3(\mathscr{D}_1,\mathscr{D}_2,\mathscr{I}_1)
\end{aligned}$$

defined analogously as in Example 4.8. They are in principle commuting tensor products with colimits. We shall later use them.

Remark 4.16. Replacing Λ by Σ in Definition 4.6 and Definition 4.10, we recover Ching's definitions in [Chi12, Definition 2.12]. These definitions can be generalized to n-fold Kelly products.

4.4. **Normal oplax monoidal structure.** To describe the relationship between these n-fold products, we need the following definition which generalized the definition for monoidal category.

Definition 4.17. ([Chi12]) A normal oplax monoidal category $(\mathscr{E},\mu_n,\alpha_{n,l,r},I)$ consists of the following data:

(1) A object I in $\mathscr{V}$, called the unit.

(2) A series of functors $\mu_0, \mu_1, \mu_2, \cdots$, where $\mu_n : \mathscr{E}^n \to \mathscr{E}$. The functor $\mu_0 : * \to \mathscr{E}$ takes the value I and $\mu_1 : \mathscr{E} \to \mathscr{E}$ is the identity functor.

(3) A series of natural transformations

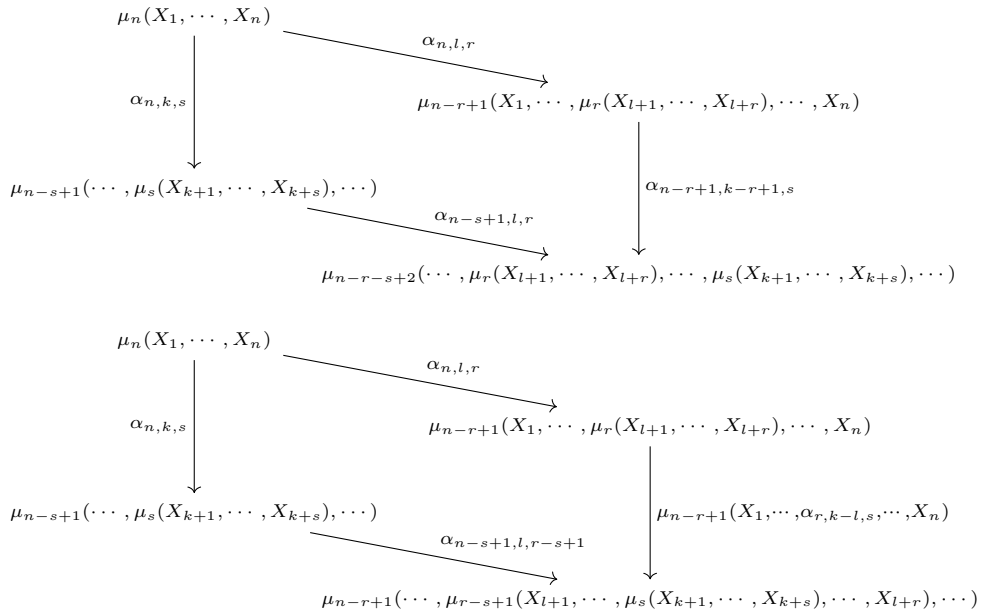
$$\alpha_{n,l,r} : \mu_n(X_1, \cdots, X_n) \to \mu_{n-r+1}(X_1, \cdots, X_l, \mu_r(X_{l+1}, \cdots, X_{l+r}), X_{l+r+1}, \cdots, X_n)$$

for triples (n, l, r) satisfying $0 \leqslant l \leqslant l + r \leqslant n$.

The data must satisfy the following conditions:

(a) For every n and l, $\alpha_{n,l,1}$ and $\alpha_{n,0,n}$ are identities;

(b) For every valid values of n, l, r, k, s, the following two diagrams commute.



A monoidal category is a normal oplax monoidal category by taking

$$\mu_n(X_1, X_2, \cdots, X_n) := (((X_1 \otimes X_2) \otimes \cdots) \otimes X_n)$$

and α to be induced by the associators. Conversely, a normal oplax monoidal structure comes from a monoidal structure if the natural transformations $\alpha_{n,l,r}$ are all isomorphisms (See [Chi12, Example 1.2]).

Example 4.18 (Reduced normal oplax monoidal structure)**.** Let $(\mathscr{E}, \mu_n, \alpha_{n,l,r}, I)$ be a normal oplax monoidal category with $\mathscr{E}$ admitting a 0-object. We define a new normal oplax monoidal structure $(\mathscr{E}, \mu_n, \overline{\alpha}_{n,l,r}, I)$ by

$$\overline{\alpha}_{n,l,r} = \begin{cases} \alpha_{n,l,r} & \text{for } r > 0 \text{ or } r = n = 0, \\ 0 & \text{for } r = 0 \text{ and } n > 0. \end{cases}$$

It is easy to see that the diagrams in (b) commute.

Example 4.19 (Truncated reduced normal oplax monoidal structure)**.** For a reduced normal oplax monoidal structure $(\mathscr{E}, \mu_n, \overline{\alpha}_{n,l,r}, I)$ as in Example 4.18 and fixed $k \geqslant 1$, we define a new normal oplax monoidal structure $(\mathscr{E}, \mu_n^{\leqslant k}, \overline{\alpha}_{n,l,r}^{\leqslant k}, I)$ by

$$\mu_n^{\leqslant k}(X_1, \cdots, X_n) = \begin{cases} \mu_n(X_1, \cdots, X_n) & \text{for } 0 \leqslant n \leqslant k, \\ 0 & \text{for } n \geqslant k + 1; \end{cases}$$

$$\overline{\alpha}^{\leqslant k}_{n,l,r} = \begin{cases} \overline{\alpha}_{n,l,r} & \text{for } 0 \leqslant n \leqslant k, \\ 0 & \text{for } n \geqslant k+1. \end{cases}$$

The diagrams in (b) still commute and this indeed forms a normal oplax monoidal structure, which we refer to as the k-th truncation of $(\mathscr{E}, \mu_n, \overline{\alpha}_{n,l,r}, I)$.

Example 4.20. From previous work we have already obtained the following data on $\Lambda^{op}[\mathscr{V}]_{\mathscr{I}_0}$:

Definitions 4.6 and 4.10	Example 4.8	Proposition 4.14	Example 4.15
μ_2, μ_3	$\alpha_{1,0,0}, \alpha_{1,1,0}$	$\alpha_{3,0,2}, \alpha_{3,1,2}$	$\alpha_{2,0,0}, \alpha_{2,1,0}, \alpha_{2,2,0}$

Recall that the following data is prescribed: μ_0 takes the value $\mathscr{I}_1$, μ_1 is the identity and $\alpha_{n,l,1}$ and $\alpha_{n,0,n}$ are the identity maps for all n, l. In summary, we have data

$$\mu_n \text{ for } n \leq 3,\ \alpha_{n,l,r} \text{ for } n \leq 2 \text{ and } \alpha_{3,l,r} \text{ for } 1 \leq r \leq 3$$

which satisfies the compatability conditions in Definition 4.17. In Theorem 4.30, we will show that the data can be extended to a normal oplax monoidal structure.

Definition 4.21. Let $(\mathscr{E}, \mu_n, \alpha_{n,l,r}, I)$ and $(\mathscr{F}, \nu_n, \beta_{n,l,r}, J)$ be two normal oplax monoidal categories. A morphism (F, t) from $\mathscr{E}$ to $\mathscr{F}$ is a functor $F : \mathscr{E} \to \mathscr{F}$ with natural transformations $t_n : F \circ \mu_n \to \nu_n \circ F^n$ such that

(1) F sends I to J.
(2) t_n is the identity for $n \leq 1$.
(3) The following diagram commutes for all (n, l, r) satisfies $0 \leqslant l \leqslant l + r \leqslant n$:

$$\begin{array}{ccc}
F\mu_n(X_1, \cdots, X_n) & \xrightarrow{F\alpha_{n,l,r}} & F\mu_{n-r+1}(X_1, \cdots, \mu_r(X_{l+1}, \cdots, X_{l+r}), \cdots, X_n) \\
 & & \downarrow{\scriptstyle t_{n-r+1}(X_1, \cdots, \mu_r(X_{l+1}, \cdots, X_{l+r}), \cdots, X_n)} \\
\downarrow{\scriptstyle t_n(X_1, \cdots, X_n)} & & \nu_{n-r+1}(FX_1, \cdots, F\mu_r(X_{l+1}, \cdots, X_{l+r}), \cdots, FX_n) \\
 & & \downarrow{\scriptstyle \nu_{n-r+1}(FX_1, \cdots, t_r(X_{l+1}, \cdots, X_{l+r}), \cdots, FX_n)} \\
\nu_n(FX_1, \cdots, FX_n) & \xrightarrow{\beta_{n,l,r}} & \nu_{n-r+1}(FX_1, \cdots, \mu_r(FX_{l+1}, \cdots, FX_{l+r}), \cdots, FX_n).
\end{array}$$

We say F is a normal oplax monoidal functor if there exist the natural transformations t_n completing it to a morphism of normal oplax monoidal categories.

Definition 4.22. Let $(\mathscr{E}, \mu, I)$ be a normal oplax monoidal category. A monoid in $\mathscr{E}$ is an object M with morphisms $m_n : \mu_n(M, M, \cdots, M) \to M$ such that the following diagram commutes for all (n, l, r) satisfying $0 \leqslant l \leqslant l + r \leqslant n$:

$$\begin{array}{ccc}
\mu_n(M, \cdots, M) & \xrightarrow{\alpha_{n,l,r}} & \mu_{n-r+1}(M, \cdots, \mu_r(M, \cdots, M), \cdots, M) \\
 & & \downarrow{\scriptstyle m_{n-r+1}(\cdots, m_r, \cdots)} \\
 & \searrow^{m_n} & M
\end{array}$$

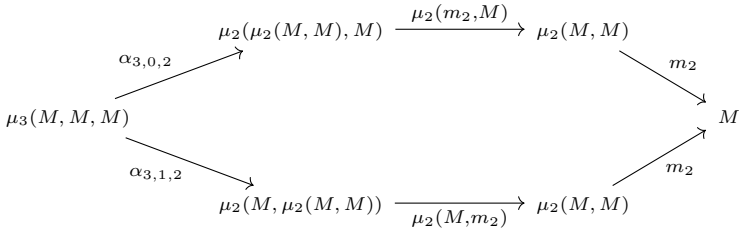
When $\mathscr{E}$ is monoidal category, the two definitions of monoids are consistent.

Lemma 4.23. *([Chi12, Prop 3.4]) Suppose we are given an object M in a normal oplax monoidal category $(\mathscr{V}, \mu_n, \alpha_{n,l,r}, I)$ with maps*

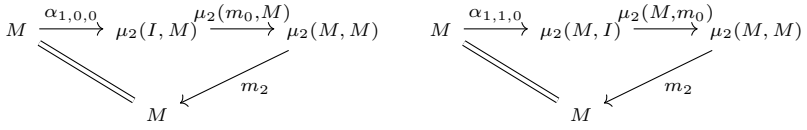
$$m_0 : I \to M \text{ and } m_2 : \mu_2(M, M) \to M.$$

They determine a monoid structure on M if and only if they satisfy the following associative laws and identity laws:

(1)



(2)



□

Motivated by this result, we make the following definition.

Definition 4.24. Let $(\mathscr{E},\mu_n,\alpha_{n,l,r},I)$ and $(\mathscr{E},\nu_n,\beta_{n,l,r},I)$ be two normal oplax monoidal structures on $\mathscr{E}$ with the same unit. We say they are *equivalent* if there is a zigzag of morphisms $(\mathrm{id},t^i):(\mathscr{E},\mu_n^i)\to(\mathscr{E},\nu_n^i)$ and $(\mathrm{id},s^i):(\mathscr{E},\mu_n^{i+1})\to(\mathscr{E},\nu_n^i)$ for $0\leq i\leq k$ with $\mu_n^0=\mu_n$ and $\mu_n^{k+1}=\nu_n$ such that t_n^i and s_n^i are all identities for $0\leq i\leq k$ and $n\leq 3$.

Corollary 4.25. *Equivalent normal oplax monoidal categories have isomorphic categories of monoids.*

Proof. It suffices to prove that $(\mathrm{id},t):(\mathscr{E},\mu_n,\alpha_{n,l,r},I)\to(\mathscr{E},\nu_n,\beta_{n,l,r},I)$ with $t_n=\mathrm{id}$ for $n\leq 3$ induces $\mathrm{Mon}(\mathscr{E},\nu_n)\cong\mathrm{Mon}(\mathscr{E},\mu_n)$. This is true because of Lemma 4.23. □

In a monoidal category, higher multiplications are completely determined by the multiplication of two objects. For normal oplax monoidal categories this is no longer the case. For example, the truncated reduced normal oplax monoidal structure $(\mathscr{E},\mu_n^{\leqslant k},\overline{\alpha}_{n,l,r}^{\leqslant k},I)$ defined in Example 4.19 are equivalent but not isomorphic for all $k\geqslant 3$.

4.5. **Lifting normal oplax monoidal structures.** In this subsection, we fix a regular cardianl $\kappa>\omega$ and use small to mean κ-small, as all diagrams considered will be κ-small.

Recall that in Theorem 3.9 we constructed a monoidal localization

$$L:\mathscr{V}\to L\mathscr{V},$$

such that $\otimes$ commutes with colimits in $L\mathscr{V}$. Therefore, $(\Lambda^{op}[L\mathscr{V}]_{\mathscr{I}_0},\odot_\Lambda,\mathscr{I}_1)$ is a monoidal category. We will prove:

Theorem 4.26. *There exists a universal normal oplax monoidal structure*

$$(\Lambda^{op}[\mathscr{V}]_{\mathscr{I}_0},\mu_n,\alpha_{n,l,r},\mathscr{I}_1)$$

extending the data given in Example 4.20, such that any extension to normal oplax monoidal structure $(\Lambda^{op}[\mathscr{V}]_{\mathscr{I}_0},\nu_n,\beta_{n,l,r},\mathscr{I}_1)$ admits a morphism to $(\Lambda^{op}[\mathscr{V}]_{\mathscr{I}_0},\mu_n)$.

Moreover, $L : (\Lambda^{op}[\mathscr{V}]_{\mathscr{I}_0}, \mu_n) \to (\Lambda^{op}[L\mathscr{V}]_{\mathscr{I}_0}, \odot_\Lambda)$ *is a functor of normal oplax monoidal categories.*

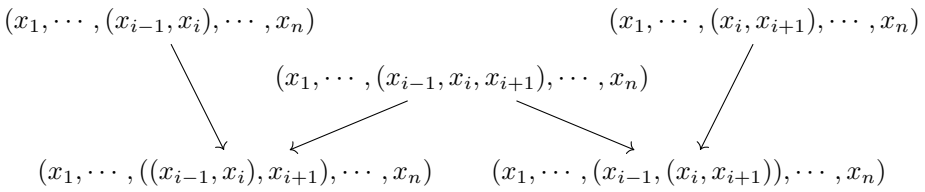
Definition 4.27. For $n \geq 3$, we define W_n to be the poset of all ways to combine n letters into one with parentheses such that each parenthesis has length of at least 2 and at most $n-1$, with $w_1 \preceq w_2$ if w_2 is obtained by adding parentheses to w_1. The length of a parenthesis is the number of factors, where parenthesized letters count as one factor.

Example 4.28. W_3 has two elements $((X_1, X_2), X_3)$ and $(X_1, (X_2, X_3))$. The following poset is part of W_4.

$$(((X_1, X_2), X_3), X_4) \longleftarrow ((X_1, X_2), X_3, X_4) \longrightarrow ((X_1, X_2), (X_3, X_4))$$

Lemma 4.29. *When* $n \geqslant 4$, W_n *is connected.*

Proof. By adding parentheses, any element in W_n is connected to some element of the form $(x_1, (\cdots, (x_i, x_{i+1}), \cdots), x_n)$ where there is a parenthesis containing two single consecutive letters. By erasing parentheses, this element is further connected to the element $(x_1, \cdots, (x_i, x_{i+1}), \cdots, x_n)$ where there are only two pairs of parenthesis. Finally, any two "adjacent" elements of this kind are connected as seen in the following diagram.



□

Theorem 4.30. *Let* $\mathscr{E}$ *be a category with small limits. Suppose that the data*

$$\mu_n \text{ for } n \leq 3,\ \alpha_{n,l,r} \text{ for } n \leq 2 \text{ or } (n = 3 \text{ and } 1 \leq r \leq 3).$$

have been defined and the conditions in Definition 4.17 are satisfied whenever the terms are defined. Then one can define μ_n *for* $n \geq 4$ *and* $\alpha_{n,l,r}$ *for the remaining cases to obtain a normal oplax monoidal structure on* $\mathscr{E}$*. Moreover, there is a terminal object* $(\mathscr{E}, \mu_n, \alpha_{n,l,r}, I)$ *in all such structures. Therefore, all such structures are equivalent in the sense of Definition 4.24.*

Proof. We define $\mu_n(X_1, \cdots, X_n)$ and $\alpha_{n,l,r}(X_1, \cdots, X_n)$ by induction on n.

Step 1. For $n \geq 4$, suppose that we have defined μ_m, $\alpha_{m,l,r}$ for $m \leq n-1$, $1 \leq r \leq m$, and $\alpha_{m,l,0}$ for $m \leq n-2$ satisfying Definition 4.17. Then for $\vec{X} = (X_1, \cdots, X_n) \in \mathscr{E}^n$, there is a diagram

$$D[\vec{X}, \mu] : W_n \to \mathscr{E}$$

sending each element to a combination of $(X_1, \cdots, X_n)$ using the corresponding $\mu_2, \cdots, \mu_{n-1}$ and each arrow to the (compositions of) suitable $\alpha_{m,l,r}$ ($m \leq n-1$, $r \geq 2$), which are all defined by inductive assumption.

Step 1.1. Define

$$\mu_n(X_1, \cdots, X_n) = \lim_{W_n} D[\vec{X}, \mu],$$

and $\alpha_{n,l,r}$ for $2 \leq r \leq n-1$ to be the canonical map

$$\mu_n(X_1, \cdots, X_n) \to D[\vec{X}, \mu](x_1, \cdots, x_l, (x_{l+1}, \cdots, x_{l+r}), x_{l+r+1}, \cdots, x_n).$$

Define $\alpha_{n,l,1}$ and $\alpha_{n,0,n}$ to be identities.

Step 1.2. For $\alpha_{n-1,l,0}$, consider

$$\vec{X}^\circ = (X_1, \cdots, X_{n-1}) \in \mathscr{E}^{n-1} \text{ and } \vec{X} = (X_1, \cdots, X_l, I, X_{l+1}, \cdots, X_{n-1}) \in \mathscr{E}^n.$$

To obtain $\alpha_{n-1,l,0} : \mu_{n-1}(\vec{X}^\circ) \to \mu_n(\vec{X})$, it suffices to see that there are canonical maps $\alpha_{n-1,l,0}(w)$ from $\mu_{n-1}(\vec{X}^\circ)$ to $D[\vec{X}](w)$ for each element $w \in W_n$ that are compatible for all w. For an element w, there are two cases:

- When there exists a parenthesis of length $n-1$ after deleting x_{l+1} from w. As $n \geq 4$, the only possiblities are $w = (x_1, (x_2, \cdots, x_n))$, $l = 0$, or $w = ((x_1, \cdots, x_{n-1}), x_n)$, $l = n-1$ or $w = (x_1, \cdots, (x_{l+\epsilon}, x_{l+\epsilon+1}), \cdots, x_n)$, $\epsilon = 0$ or 1. We let $\alpha_{n-1,l,0}(w)$ be

  $$\alpha_{1,0,0} : \mu_{n-1}(\vec{X}^\circ) \to \mu_2(I, \mu_{n-1}(\vec{X}^\circ))$$

  in the first case,

  $$\alpha_{1,1,0} : \mu_{n-1}(\vec{X}^\circ) \to \mu_2(\mu_{n-1}(\vec{X}^\circ), I)$$

  in the second case, and $\mu_{n-1}(X_1, \cdots, \alpha_{2,\epsilon,0}, \cdots, X_{n-1}) \circ \alpha_{n-1,l+\epsilon-1,1}$ in the last case.
- When there are no parenthesis of length $n-1$ after deleting x_{l+1} from w. We obtain an element $w^\circ \in W_{n-1}$ by deleting x_{l+1} and any resulting parathesis of length 1. Let $\alpha_{n-1,l,0}(w)$ be

  $$\mu_{n-1}(\vec{X}^\circ) \to D[\vec{X}^\circ](w^\circ) \to D[\vec{X}](w),$$

  where the first map is some $\alpha_{3,l',2}$ when $n = 4$ and the canonical map from the limit when $n > 4$, and the second map is induced by some $\alpha_{n',l',0}$ for $n' \leq n-2$, possibly composed with some $\alpha_{n'',l'',1}$ for $n'' \leq n-2$ if there have been a parathesis of length 1 deleted.

The compactiblity of the $\alpha_{n-1,l,0}(w)$ across w holds by inductive hypothesis because all μ_m appearing in the diagrams satsifies $m \leq n-1$.

Step 1.3. We still need to check the conditions in Definition 4.17. There are essentially three cases in Definition 4.17(b):

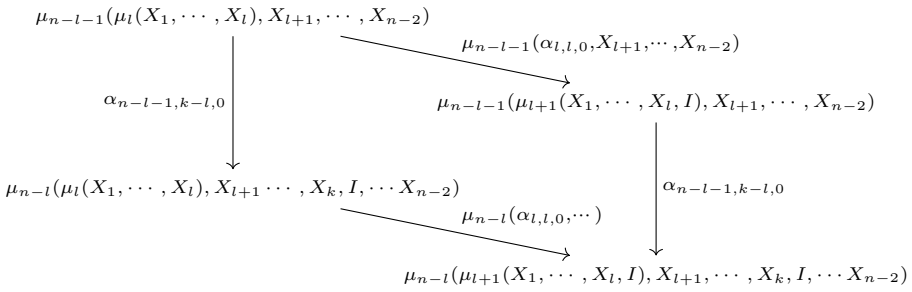
(1) $s, r \neq 0$, we should check for $\mu_n(X_1, \cdots, X_n)$.
(2) $s = 0$, $r \neq 0$, we should check for $\mu_{n-1}(X_1, \cdots, X_{n-1})$.
(3) $s = r = 0$, we should check for $\mu_{n-2}(X_1, \cdots, X_{n-2})$.

The diagrams in case (1) commute by definition of μ_n. The diagrams in case (2) commute by definition of $\alpha_{n-1,l,0}$. For case (3), let

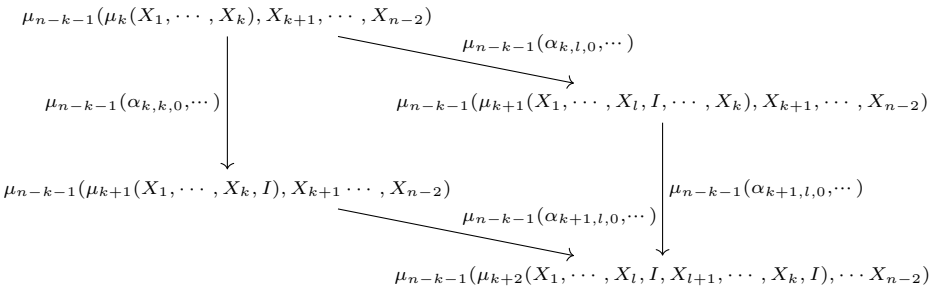
$$\vec{X}^\circ = (X_1, \cdots, X_{n-2}) \text{ and } \vec{X} = (X_1, \cdots, X_l, I, \cdots, X_k, I, \cdots, X_{n-2}) \in \mathscr{E}^n.$$

We need to show that the two composites $\alpha_{n-1,k+1,0} \circ \alpha_{n-2,l,0}$ and $\alpha_{n-1,l,0} \circ \alpha_{n-2,k,0}$ give the same map $\mu_{n-2}(\vec{X}^\circ) \to \mu_n(\vec{X})$. For this purpose, it suffices to check that they give the same map $\mu_{n-2}(\vec{X}^\circ) \to D[\vec{X}, \mu](w)$ for each $w \in W_n$. We may assume that $w = (x_1, \cdots, (x_i, \cdots, x_j), \cdots, x_n)$ has two pairs of parantheses, as any w receives a map from such an element. Work similarly as before, we attempt to delete the letters x_{l+1} and x_{k+2} whose values are I in $\vec{X}$ from w and there are several cases:

- When exactly one of x_{l+1} and x_{k+2} is in the inner paranthesis. Without loss of generality we do it for $w = ((x_1, \cdots, x_{l+1}), \cdots, x_n)$. Then the following diagram commutes by naturality of $\alpha_{n-l-1,k-l,0}$:



- When both of x_{l+1} and x_{k+2} are in the inner paranthesis, or none of them is in the inner paranthesis. Without loss of generality we do it for $w = ((x_1, \cdots, x_{l+1}, \cdots, x_{k+2}), \cdots, x_n)$. Then the following diagram commutes by inductive hypothesis:



Step 2. We now check the universal property. Let $(\mathscr{E}, \nu_n, \beta_{n,l,r}, I)$ be a normal oplax monoidal structure with $\nu_n = \mu_n$ and $\beta_{n,l,r} = \alpha_{n,l,r}$ for $n \leq 3$. We show there is a morphism s_n from this normal oplax structure to $(\mathscr{E}, \mu_n, \alpha_{n,l,r}, I)$. When $n \leqslant 3$, s_n is the identity. Suppose we have constructed s_m for all $m \leqslant n-1$, these s_m induce natural morphisms from $D[\vec{X}, \nu]$ to $D[\vec{X}, \mu]$. Therefore we have natural maps between their limits. On the other hand, the natural transformations $\beta_{m,l,r}$ for $m \leq n$ induce compatible maps from $\nu_n(X_1, \cdots, X_n)$ to each entry of the diagram $D[\vec{X}, \nu]$, thus to its limit. We set s_n to be the dotted arrow.

$$\begin{array}{ccc} \nu_n(X_1,\cdots,X_n) & \longrightarrow & \lim_{W_n} D[\vec{X},\nu] \\ {\scriptstyle s_n}\vdots & & \downarrow \\ \mu_n(X_1,\cdots,X_n) & = & \lim_{W_n} D[\vec{X},\mu] \end{array}$$

□

Proof of Theorem 4.26. By Theorem 4.30, we have a terminal normal oplax monoidal structure $(\Lambda^{op}[\mathscr{V}]_{\mathscr{I}_0}, \mu_n, \alpha_{n,l,r}, \mathscr{I}_1)$ extending the datum in Example 4.20. In particular, by definition there is

$$\mu_n(\mathscr{D}_1, \cdots, \mathscr{D}_n) := \lim_{W_n} D[\vec{\mathscr{D}}, \mu],$$

where limit in $\Lambda^{op}[\mathscr{V}]_{\mathscr{I}_0}$ is taken pointwise. We show that $L : \Lambda^{op}[\mathscr{V}]_{\mathscr{I}_0} \to \Lambda^{op}[L\mathscr{V}]_{\mathscr{I}_0}$ is a functor of normal oplax monoidal categories.

As $(\Lambda^{op}[L\mathscr{V}]_{\mathscr{I}_0}, \odot_\Lambda, \mathscr{I}_0)$ is a monoidal category, it has a normal oplax monoidal structure $(\upsilon_n, \beta_{n,l,r})$ and the arrows in the diagram $D[L\vec{\mathscr{D}}, \upsilon]$ are all isomorphisms in $\Lambda^{op}[L\mathscr{V}]$, so are the arrows from $\upsilon_n(L\mathscr{D}_1, \cdots, L\mathscr{D}_n)$ to each term of the diagram. As W_n is a connected poset by Lemma 4.29, we have

$$\upsilon_n(L\mathscr{D}_1, \cdots, L\mathscr{D}_n) \cong \lim_{W_n} D[L\vec{\mathscr{D}}, \upsilon].$$

By Theorem 3.9, L preserves small limits and colimits. We have

$$L(\mu_2(\mathscr{D}_1, \mathscr{D}_2)) \cong L\mathscr{D}_1 \odot_\Lambda L\mathscr{D}_2 \text{ and } L\mu_3(\mathscr{D}_1, \mathscr{D}_2, \mathscr{D}_3) \cong L\mathscr{D}_1 \odot_\Lambda L\mathscr{D}_2 \odot_\Lambda L\mathscr{D}_3,$$

as well as natural maps

$$L\mu_n(\mathscr{D}_1, \cdots, \mathscr{D}_n) \cong \upsilon_n(L\mathscr{D}_1, \cdots, L\mathscr{D}_n) \text{ and } L\alpha_{n,l,r} \cong \beta_{n,l,r}.$$

□

Corollary 4.31. *$\Lambda^{op}_{\mathcal{I}}[\mathscr{V}] \subset \Lambda^{op}[\mathscr{V}]_{\mathscr{I}_0}$ is a sub-normal oplax monoidal category.*

Proof. By inspection, the category $\Lambda^{op}_{\mathcal{I}}[\mathscr{V}]$ is closed under operators μ_n. □

4.6. **Unital operads as monoids.** We are now ready to proof the following theorem, generalizing [MZZ25, Theorem 0.13].

Theorem 4.32. *Let $\mathscr{V}$ be a complete and cocomplete symmetric monoidal category. Let $\Lambda^{op}[\mathscr{V}]_{\mathcal{I}_0}$ and $\Lambda^{op}_{\mathcal{I}}[\mathscr{V}]$ be equipped with the normal oplax monoidal structures defined in Theorem 4.26. There are isomorphisms of categories*

$$\{\textit{based operads in } \mathscr{V}\} \cong \{\textit{monoids in } \Lambda^{op}[\mathscr{V}]_{\mathscr{I}_0}\};$$
$$\{\textit{unital operads in } \mathscr{V}\} \cong \{\textit{monoids in } \Lambda^{op}_{\mathcal{I}}[\mathscr{V}]\}.$$

Proof. It is easy to see that given a monoid in $\Lambda^{op}[\mathscr{V}]_{\mathcal{I}_0}$ (or respectively, $\Lambda^{op}_{\mathcal{I}}[\mathscr{V}]$), the structure maps m_0 and m_2 provide the structure maps of a based (respectively a unital) operad on $\mathscr{V}$. Conversely, suppose that we are given a based (unital) operad $\mathscr{C}$ in $\mathscr{V}$. The operadic action γ and unit η gives the desired structure map $m_2 \colon \mathscr{C} \odot_\Lambda \mathscr{C} \to \mathscr{C}$ and $m_0 \colon \mathscr{I}_1 \to \mathscr{C}$. It suffices to show these maps satisfy the compatibilities in Lemma 4.23. By Definition 4.6 and Definition 4.10, μ_2 and μ_3 are colimits of terms $\mathscr{C}(\mathbf{k}) \otimes \bigotimes \mathscr{C}(\mathbf{n}_r)$ and $\mathscr{C}(\mathbf{k}) \otimes \bigotimes \mathscr{C}(\mathbf{n}_r) \otimes \bigotimes \mathscr{C}(\mathbf{j}_t)$. Termwisely these compatibilities are precisely the assciative, left and right unit laws. This proves that $\mathscr{C}$ is a monoid in $\Lambda^{op}[\mathscr{V}]_{\mathscr{I}_0}$ (respectively $\Lambda^{op}_{\mathcal{I}}[\mathscr{V}]$). □

Remark 4.33. As we see in Theorem 4.26, all normal oplax monoidal structures satisfying the requirements are equivalent in the sense of Definition 4.24. Consequently, the category of operads is isomorphic to the category of monoids in Λ-sequences equipped with *any* normal oplax monoidal structure by Corollary 4.25.

Remark 4.34. Ching [Chi12] constructed a normal oplax monoidal structure on Σ-sequences, denoted $(\Sigma^{op}[\mathscr{V}], \mu_n, \alpha_{n,l,r}, \mathscr{I}_1^\Sigma)$ here, to prove an analogue of Theorem 4.32 identifying operads in $\mathscr{V}$ to monoids in $\Sigma^{op}[\mathscr{V}]$. Our constructions in this section can be applied to Σ-sequences similarly to obtain another normal oplax monoidal structure $(\Sigma^{op}[\mathscr{V}], \nu_n, \beta_{n,l,r}, \mathscr{I}_1^\Sigma)$. Ching's structure has a morphism to our structure, since it satisfies the conditions in Theorem 4.26. We do not know whether they are isomorphic.

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